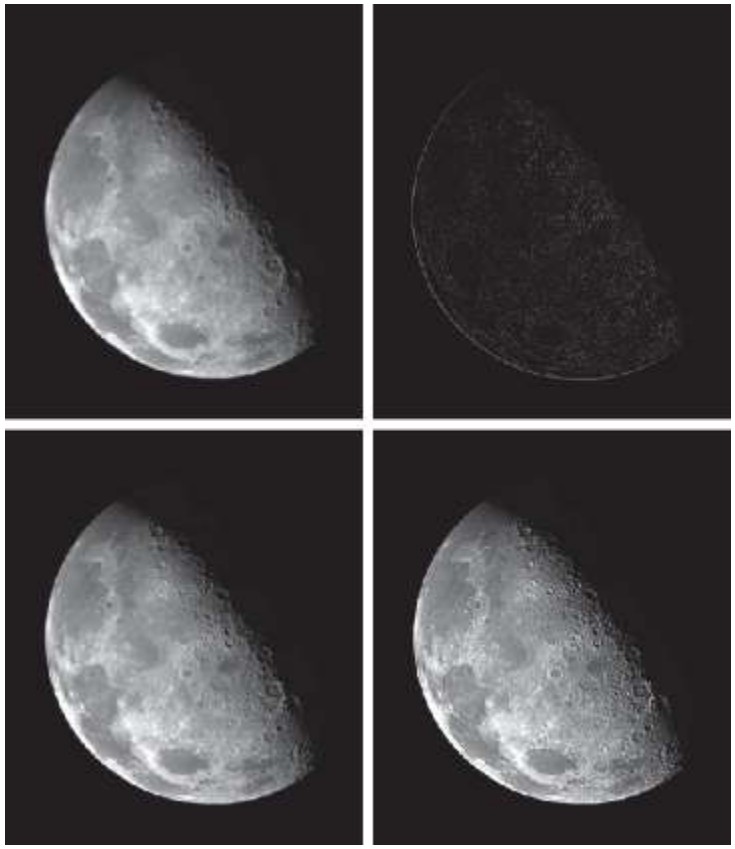
Image Processing – CH3



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Intensity Transformation
Spatial Filtering

- **The spatial domain:**
  - The image plane: in a Cartesian coordinate system of discrete rows and columns.
  - At the intersection of each row and column is a pixel.
  - Each pixel has a value, which we will call intensity.
- **The frequency domain :**
  - A (2-dimensional) discrete transform of the spatial domain

- **Transformation and Filtering :**
  - To “improve” the usefulness of an image by using some transformation on the image.
  - Often the improvement is to help make the image “better” looking, such as increasing the intensity or contrast.

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Background

- **A mathematical representation of spatial domain process:**

$$g(x, y) = T[f(x, y)]$$

where *f*(*x*, *y*): the input image

$g(x, y)$: the processed image

***T*: an operator on *f*, defined over some neighborhood of (*x*, *y*)**

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Background

- The neighborhood of (x, y) and the predefined operation
  - Spatial filter, spatial mask, kernel, template, window

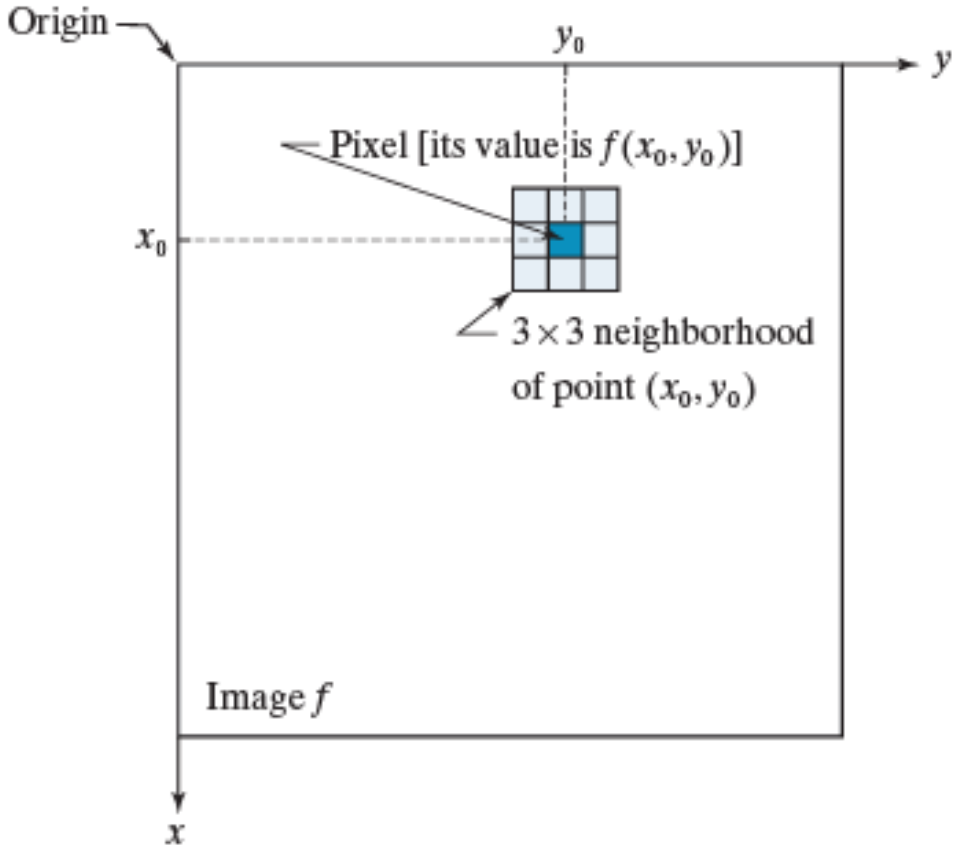

FIGURE 3.1 A 3×3 neighborhood about a point (x_0, y_0) in an image. The neighborhood is moved from pixel to pixel in the image to generate an output image. Recall from Chapter 2 that the value of a pixel at location (x_0, y_0) is $f(x_0, y_0)$, the value of the image at that location.

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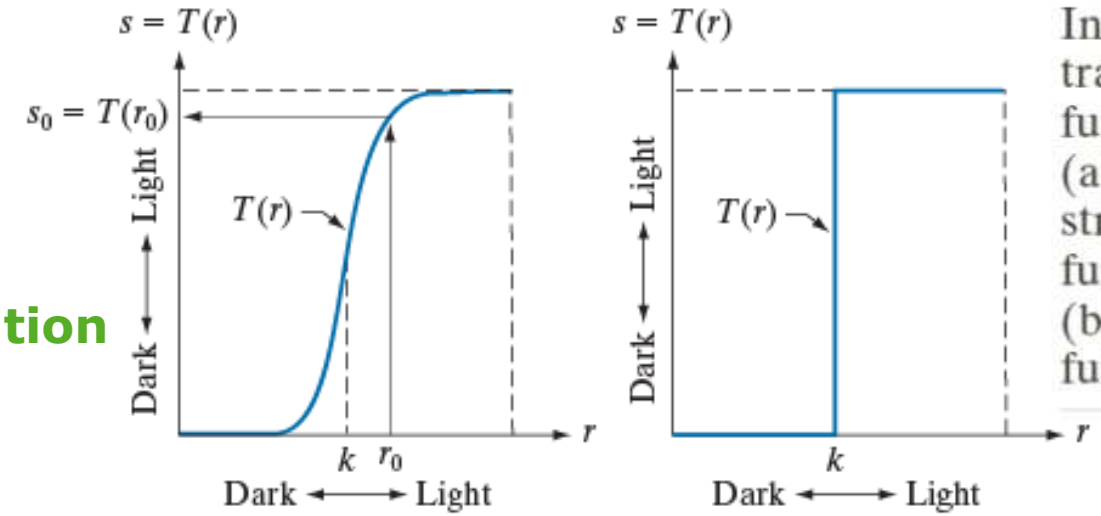
Intensity Transformation

- Point processing
- Gray-level mapping transformation



a b

FIGURE 3.2 Intensity transformation functions. (a) Contrast-stretching function. (b) Thresholding function.

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Intensity Transformation

FIGURE 3.3 Some basic intensity transformation functions. Each curve was scaled *independently* so that all curves would fit in the same graph. Our interest here is on the *shapes* of the curves, not on their relative values.

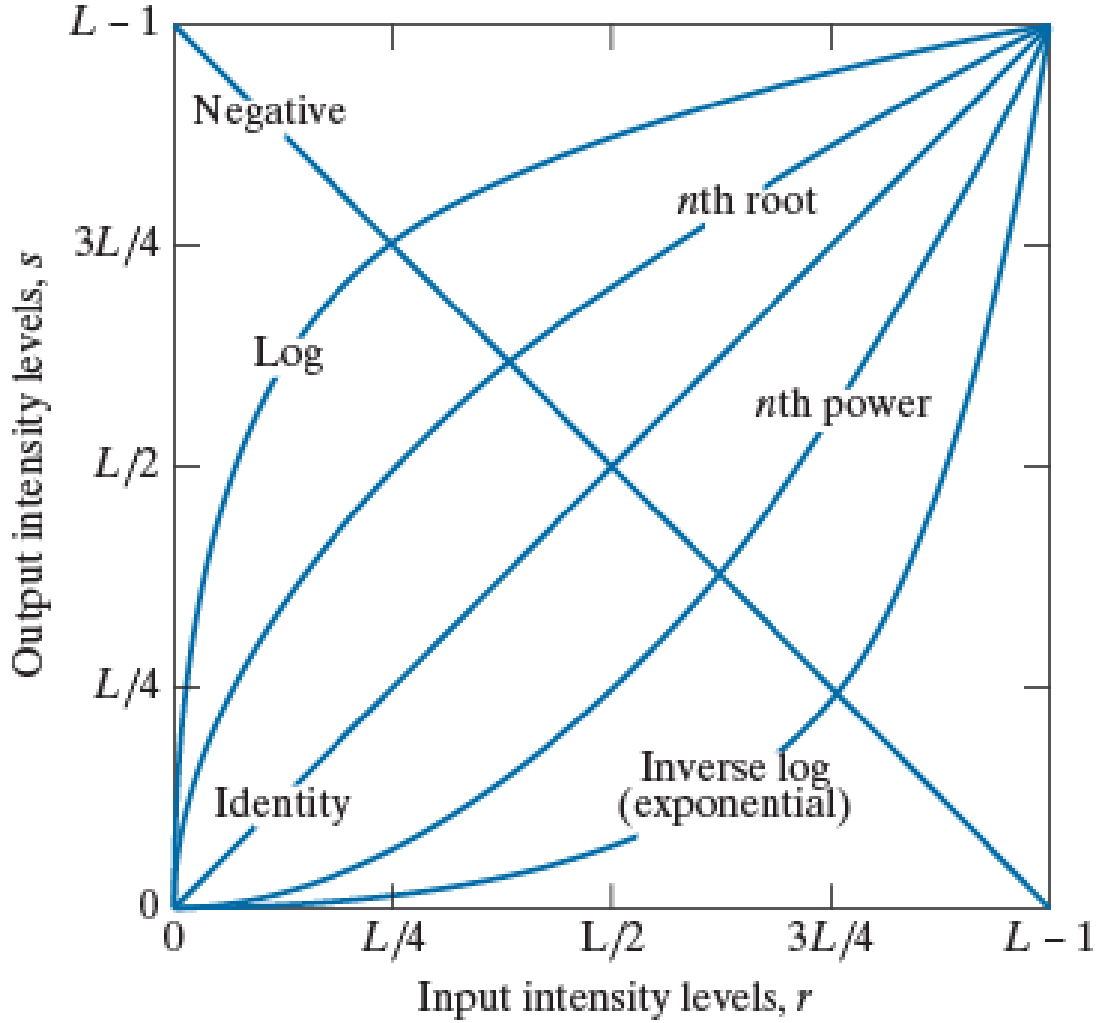

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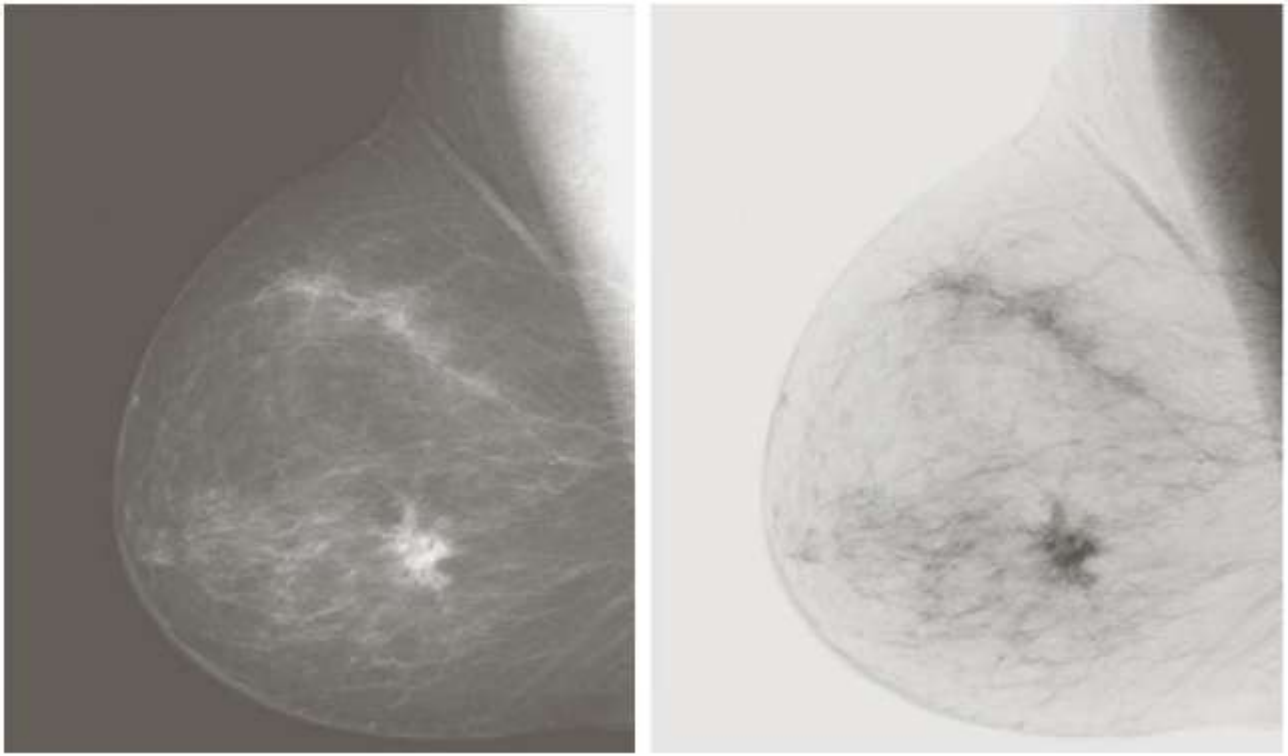
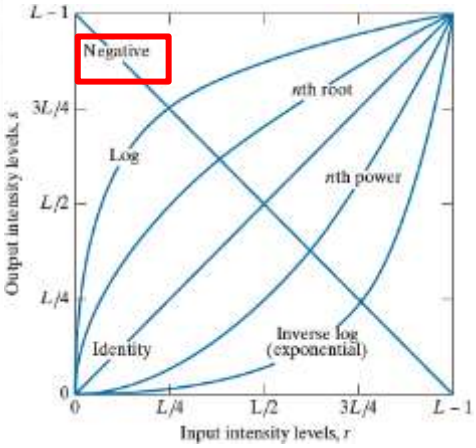
Image Negatives

- **Let the range of gray level be [0, *L*-1], then *s=L-1-r***



a b

FIGURE 3.4 (a) A digital mammogram. (b) Negative image obtained using Eq. (3-3). (Image (a) Courtesy of General Electric Medical Systems.)



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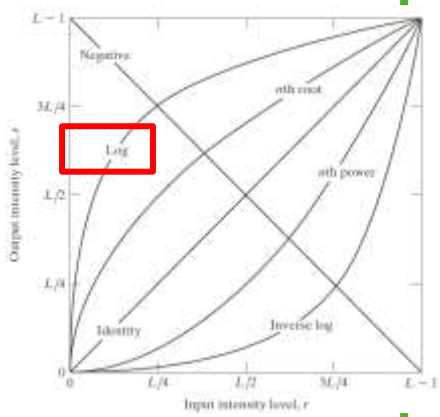
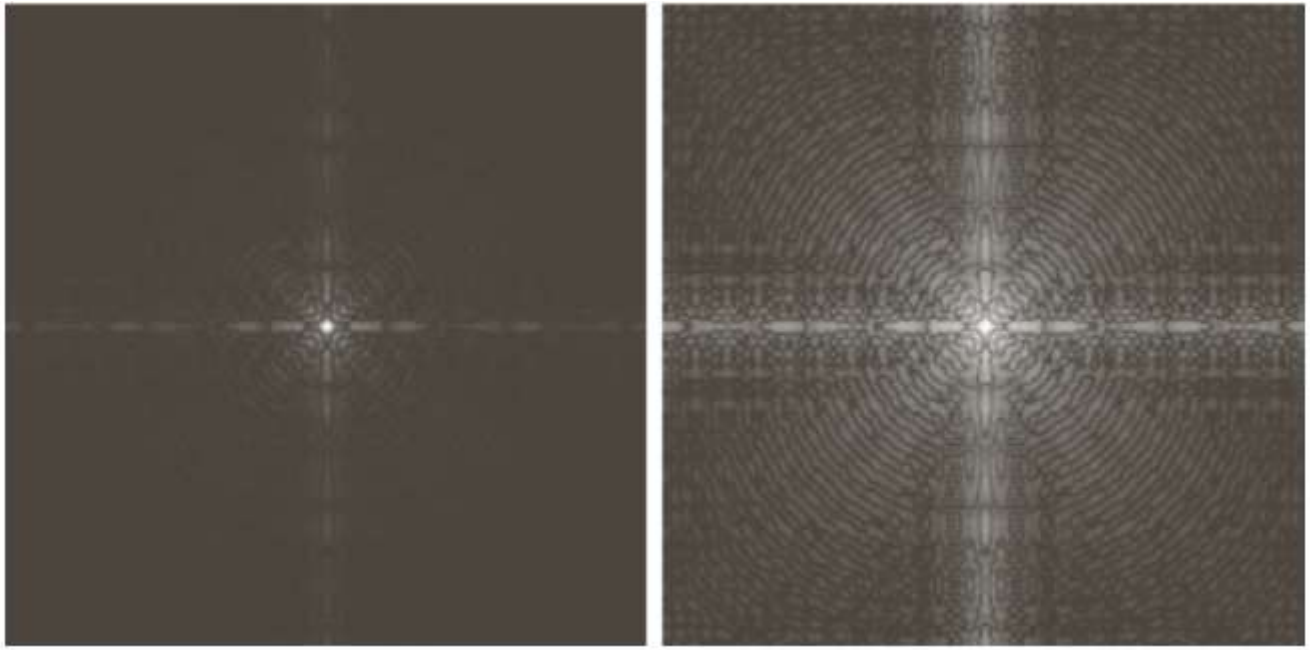
Log Transformations

- To expand the values of dark pixels in an image while compressing the higher-level values
- s=clog(1+r), where r>=0 and c is a constant





a b

FIGURE 3.5 (a) Fourier spectrum displayed as a grayscale image. (b) Result of applying the log transformation in Eq. (3-4) with $c = 1$. Both images are scaled to the range [0, 255].

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Power-Law (Gamma) Transformations

- $s=cr^{\gamma}$, c,γ are positive constants

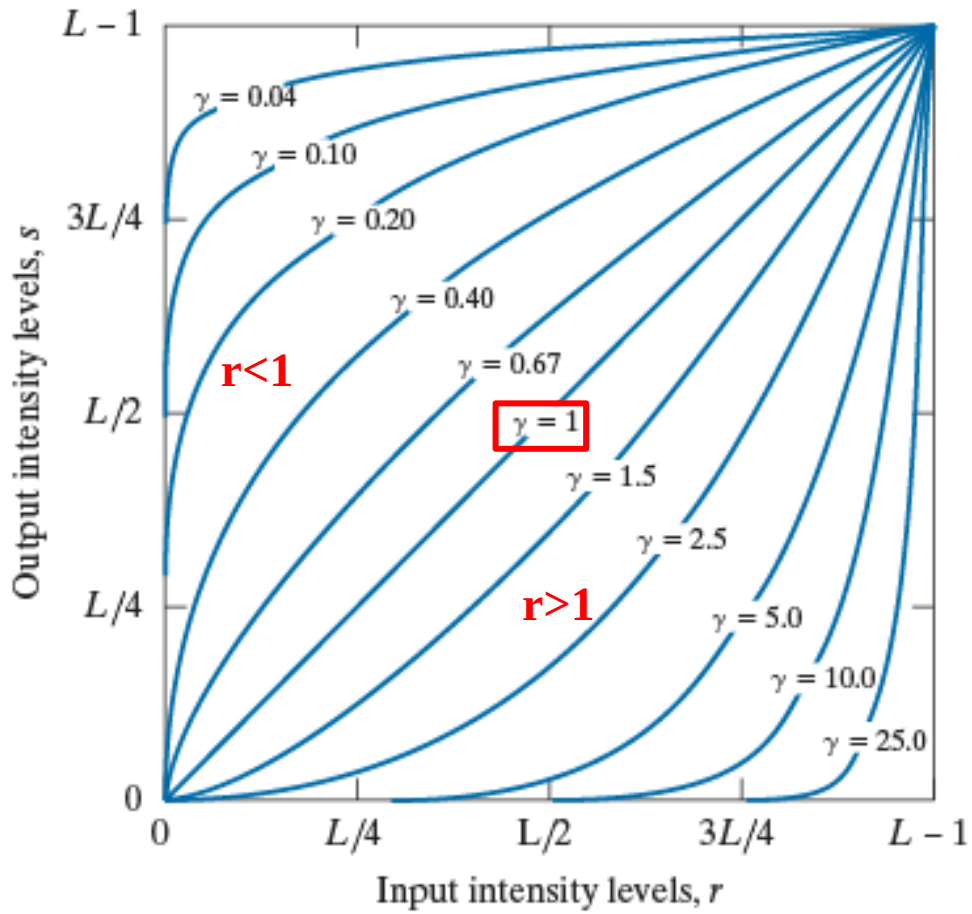

FIGURE 3.6 Plots of the gamma equation $s = cr^{\gamma}$ for various values of γ ($c = 1$ in all cases). Each curve was scaled *independently* so that all curves would fit in the same graph. Our interest here is on the *shapes* of the curves, not on their relative values.

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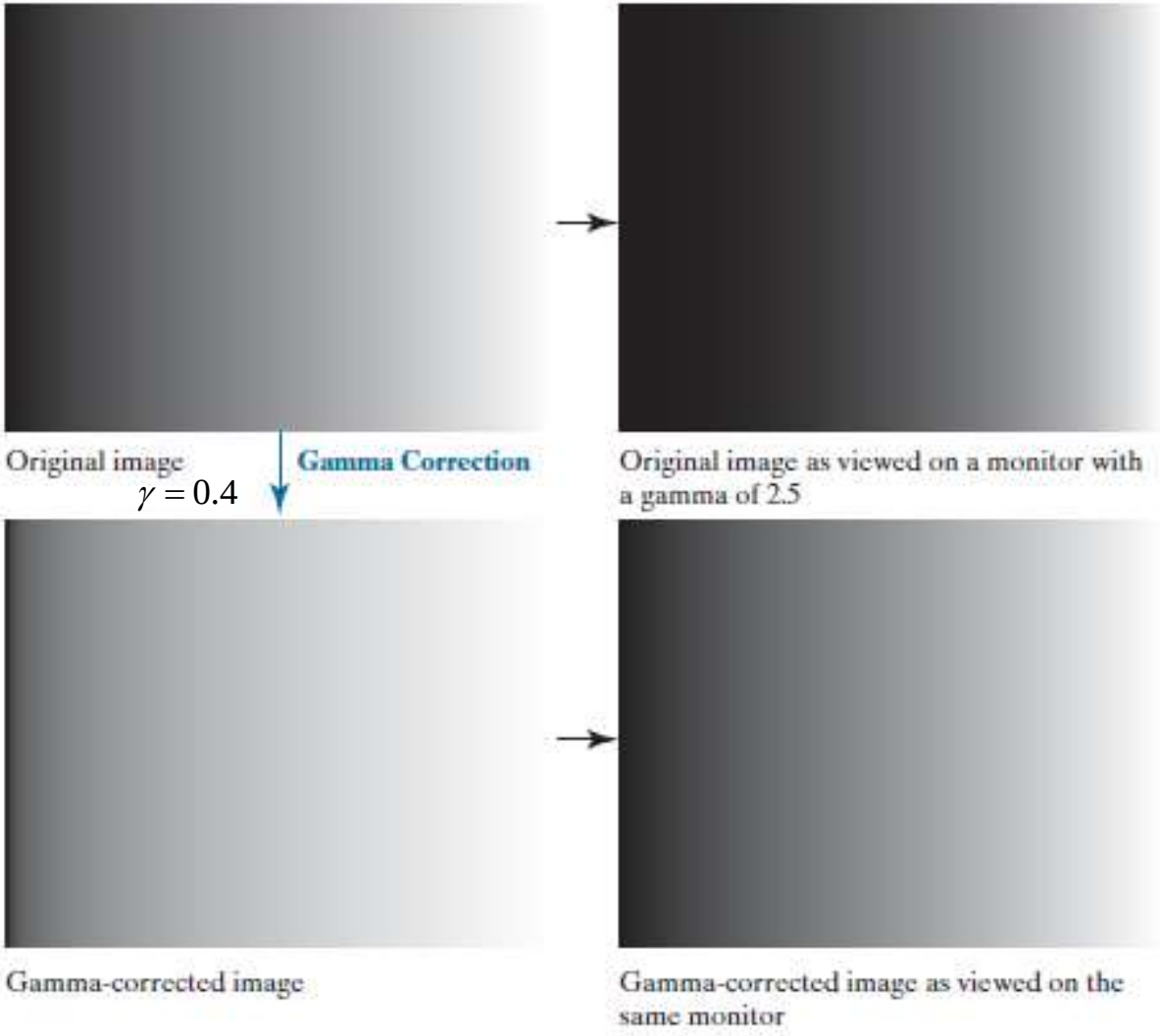
Power-Law Transformations

Gamma Correction



a b
c d

FIGURE 3.7 (a) Intensity ramp image. (b) Image as viewed on a simulated monitor with a gamma of 2.5. (c) Gamma-corrected image. (d) Corrected image as viewed on the same monitor. Compare (d) and (a).

$$s = cr^{\gamma}$$

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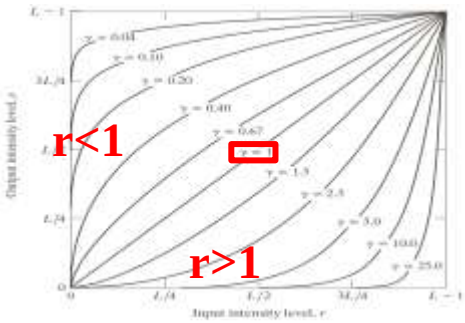
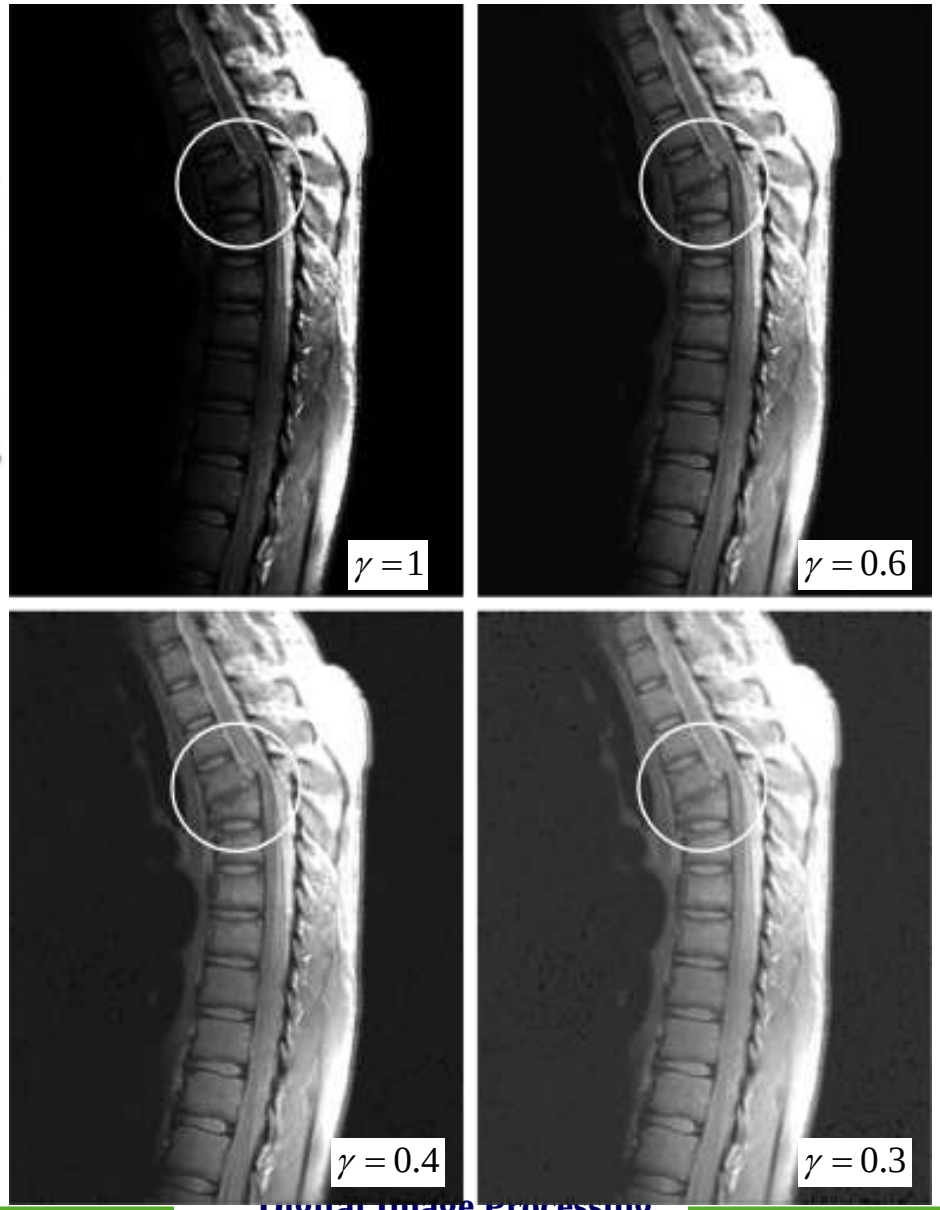
Power-Law Transformations

Gamma Correction





a b
c d

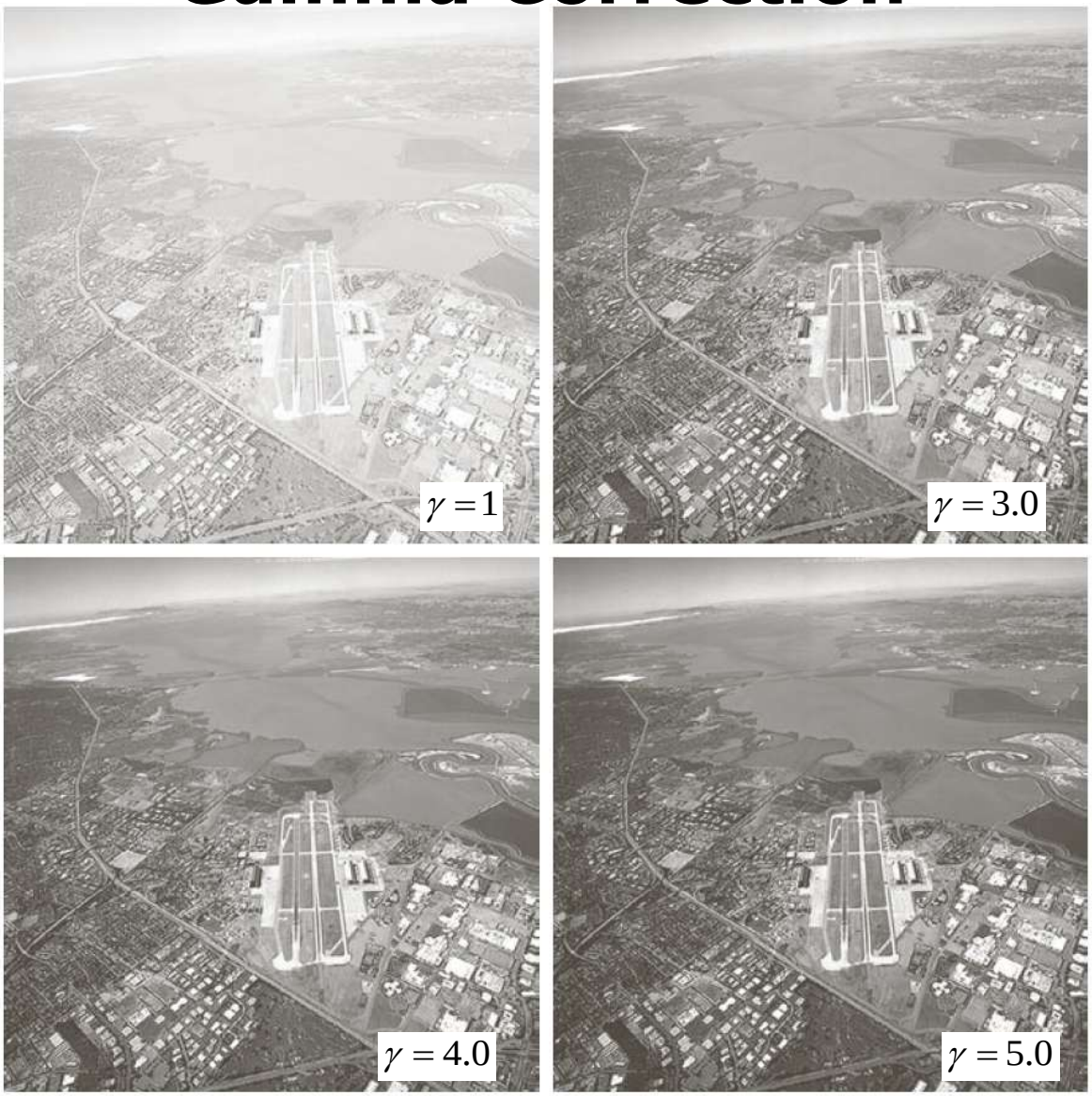
FIGURE 3.8 (a) Magnetic resonance image (MRI) of a fractured human spine (the region of the fracture is enclosed by the circle). (b)–(d) Results of applying the transformation in Eq. (3-5) with $c = 1$ and $\gamma = 0.6, 0.4$, and 0.3, respectively. (Original image courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

$$s = cr^{\gamma}$$

為保護所參考來源之智慧財產權，此教材僅為上課時教學之用。

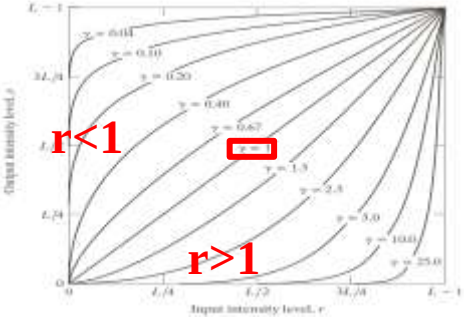
Power-Law Transformations

Gamma Correction



a b
c d

FIGURE 3.9 (a) Aerial image. (b)–(d) Results of applying the transformation in Eq. (3-5) with $\gamma = 3.0$, 4.0, and 5.0, respectively. ($c = 1$ in all cases.) (Original image courtesy of NASA.)



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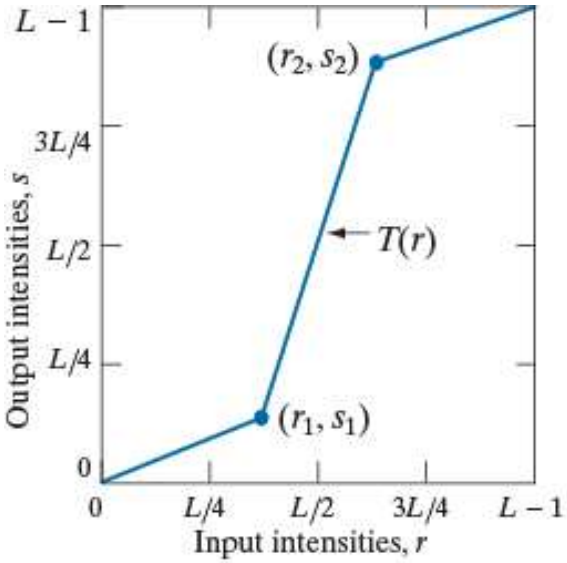
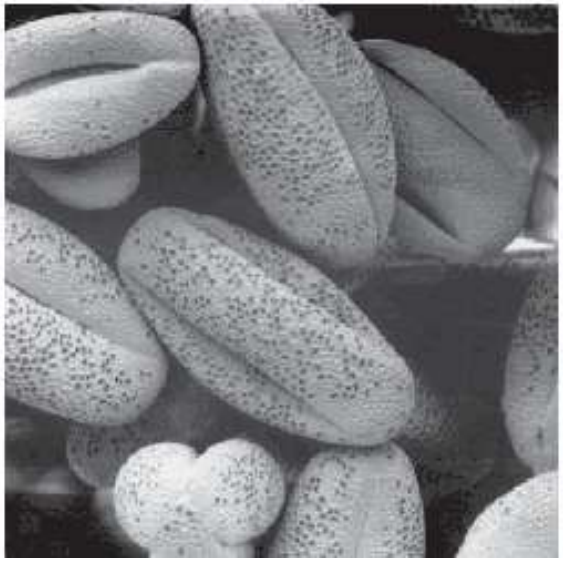
Contrast Stretching

a b
c d

FIGURE 3.10 Contrast stretching. (a) Piecewise linear transformation function. (b) A low-contrast electron microscope image of pollen, magnified 700 times. (c) Result of contrast stretching. (d) Result of thresholding. (Original image courtesy of Dr. Roger Heady, Research School of Biological Sciences, Australian National University, Canberra, Australia.)





Threshold=mean

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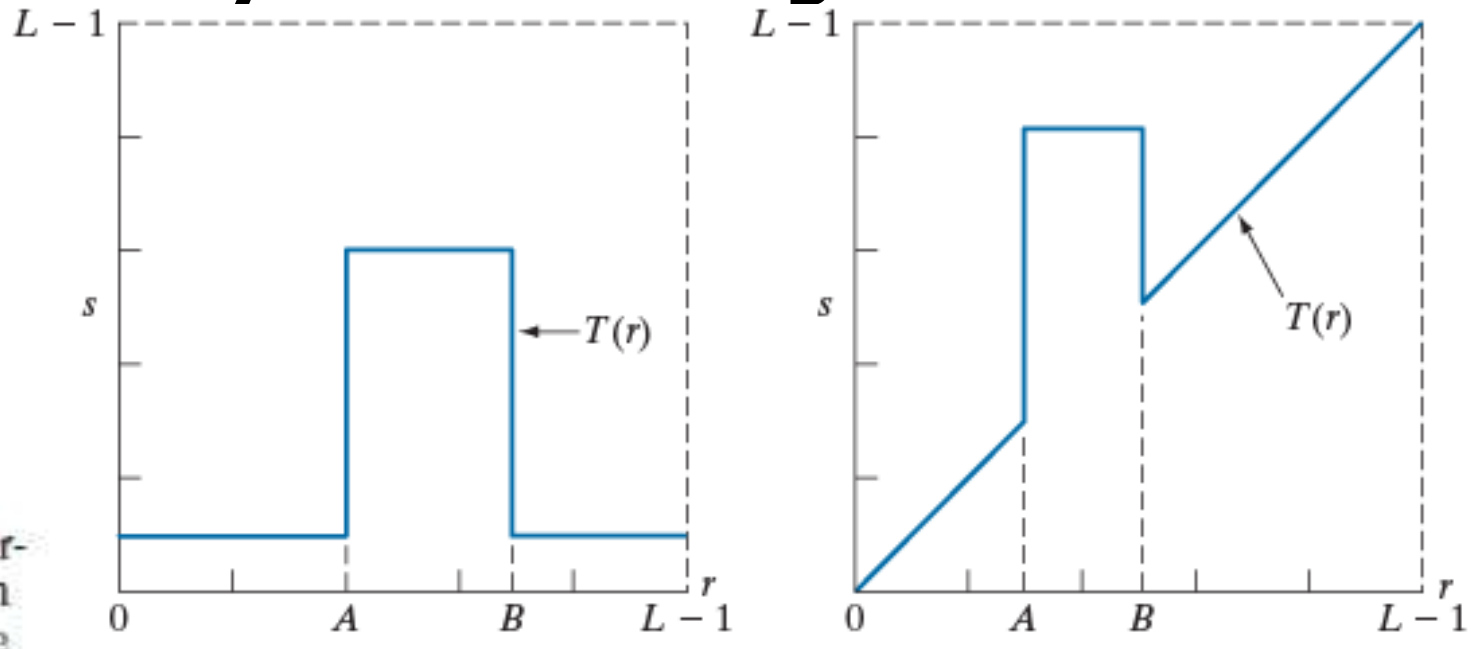
Intensity-Level Slicing

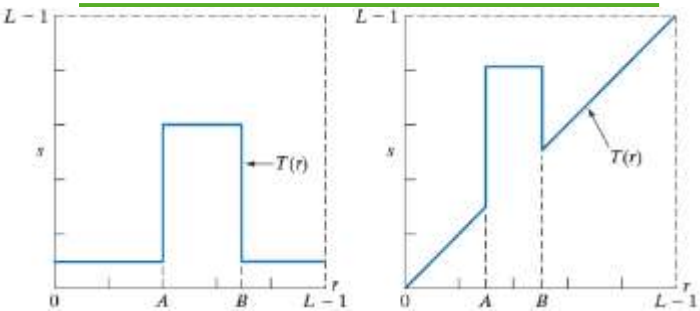


a b

FIGURE 3.11 (a) This transformation function highlights range $[A, B]$ and reduces all other intensities to a lower level. (b) This function highlights range $[A, B]$ and leaves other intensities unchanged.

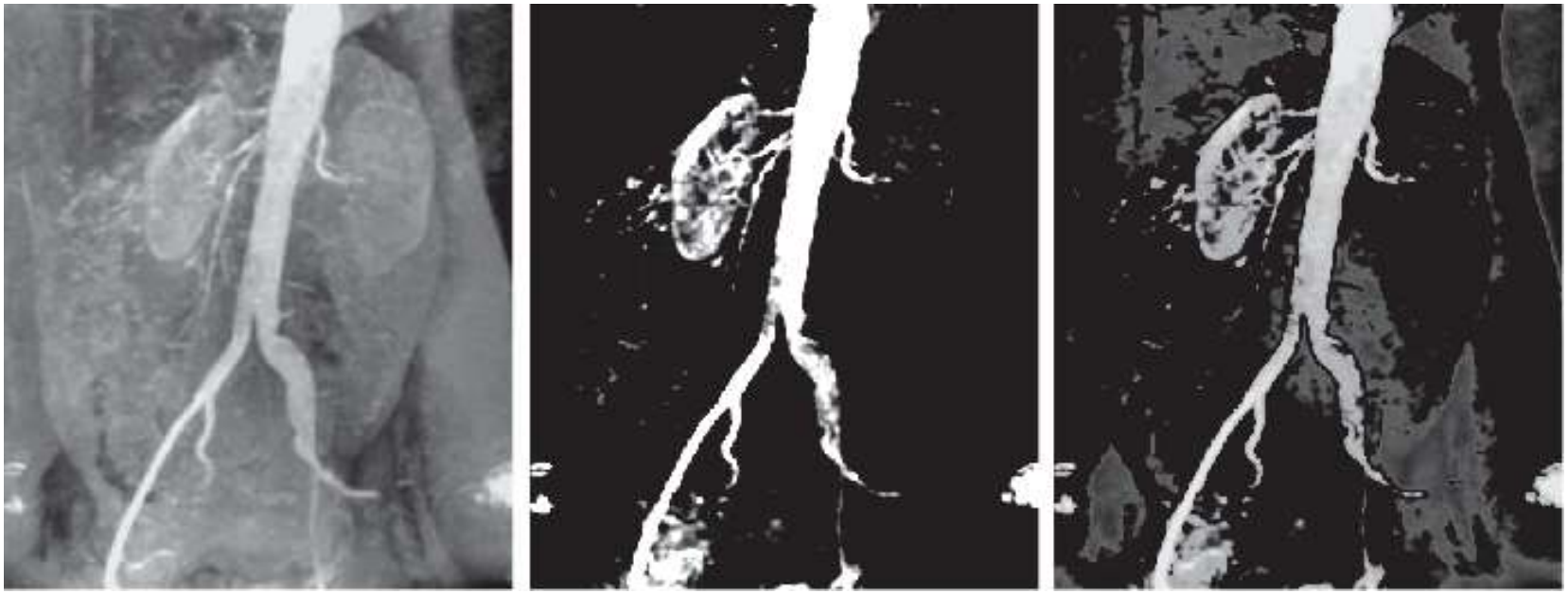
為保護所參考來源之智慧財產權，此教材僅為上課時教學之用。

Intensity-Level Slicing



a b c

FIGURE 3.12 (a) Aortic angiogram. (b) Result of using a slicing transformation of the type illustrated in Fig. 3.11(a), with the range of intensities of interest selected in the upper end of the gray scale. (c) Result of using the transformation in Fig. 3.11(b), with the selected range set near black, so that the grays in the area of the blood vessels and kidneys were preserved. (Original image courtesy of Dr. Thomas R. Gest, University of Michigan Medical School.)



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Bit-Plane Slicing

❖ Bit-plane slicing:

- It can highlight the contribution made to total image appearance by specific bits.
- Each pixel in an image represented by 8 bits.
- Image is composed of eight 1-bit planes, ranging from bit-plane 0 for the least significant bit to bit plane 7 for the most significant bit.

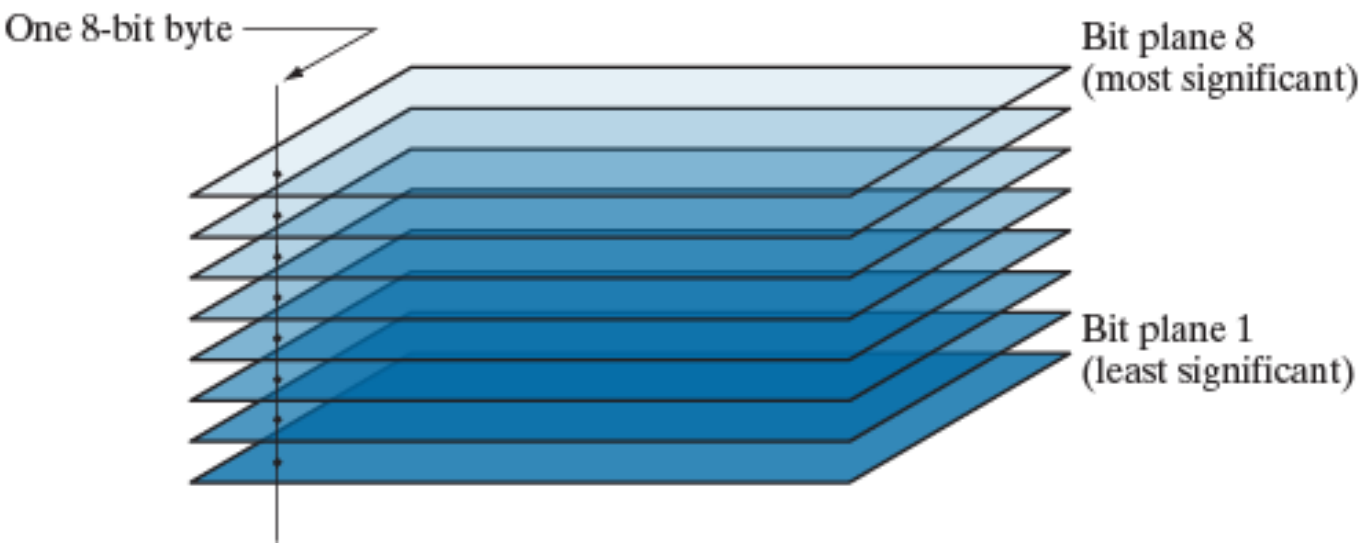

FIGURE 3.13 Bit-planes of an 8-bit image.

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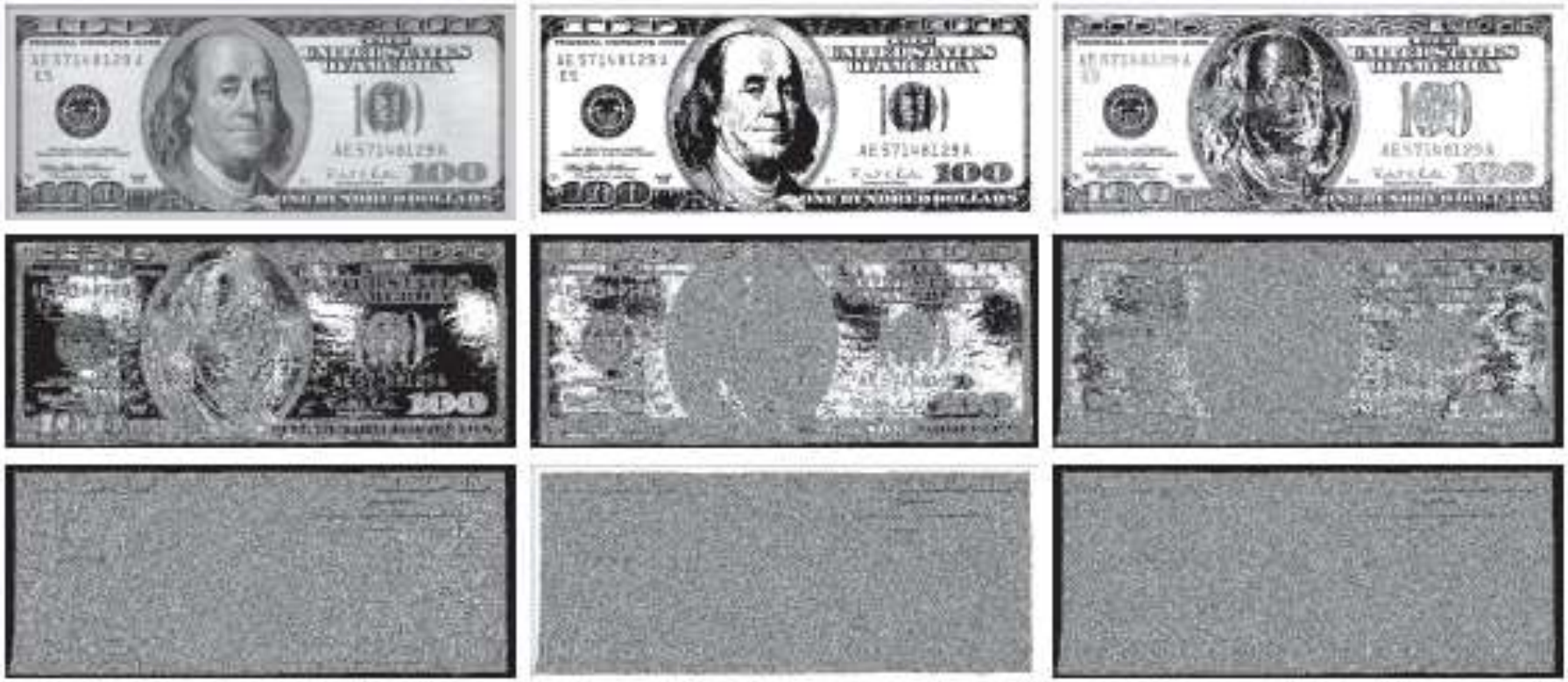
Bit-Plane Slicing



a b c
d e f
g h i

FIGURE 3.14 (a) An 8-bit gray-scale image of size 550×1192 pixels. (b) through (i) Bit planes 8 through 1, with bit plane 1 corresponding to the least significant bit. Each bit plane is a binary image..

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Bit-Plane Slicing

a b c **FIGURE 3.15** Image reconstructed from bit planes: (a) 8 and 7; (b) 8, 7, and 6; (c) 8, 7, 6, and 5.

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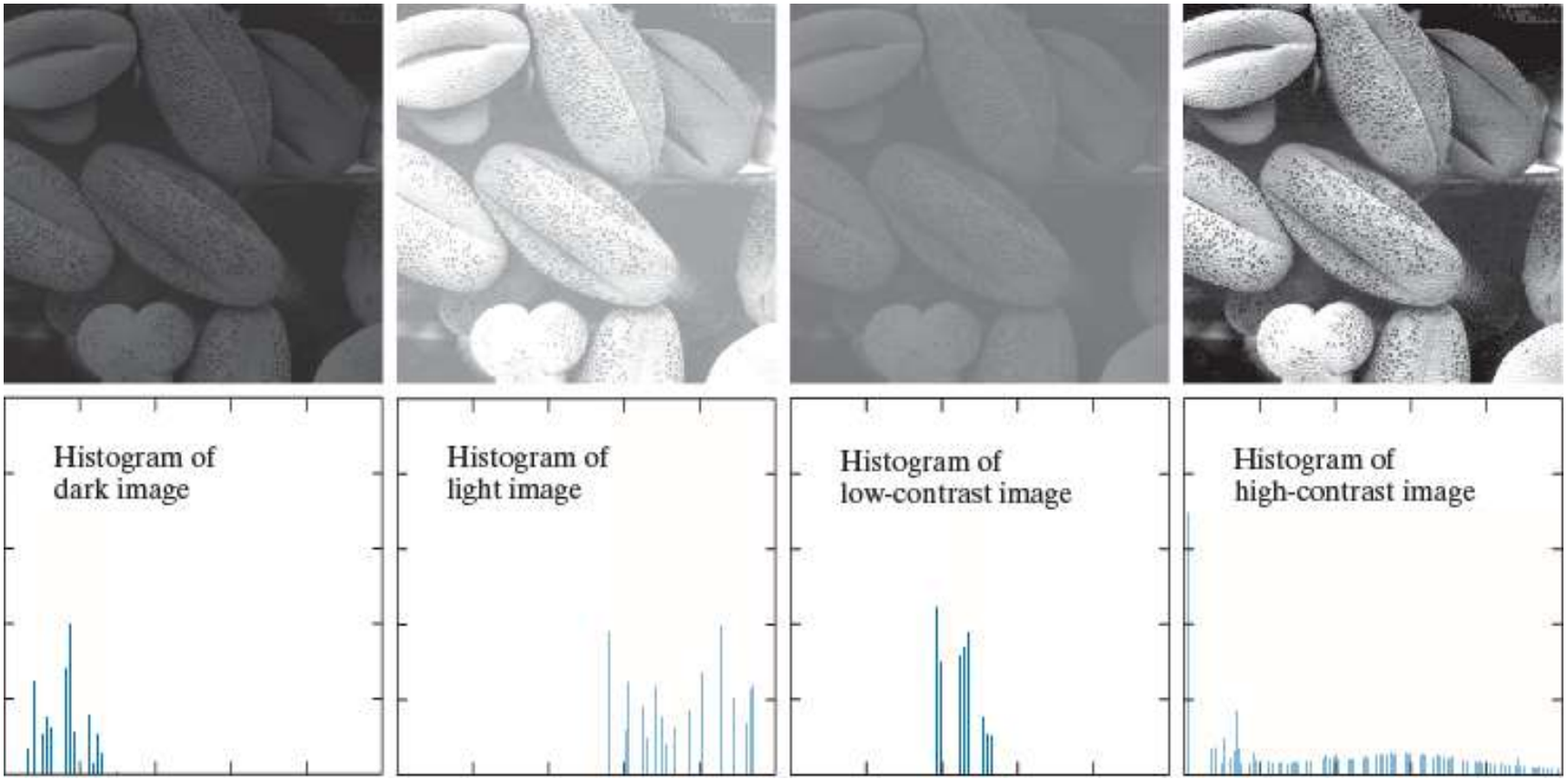
Histogram Processing

$$h(r_k) = n_k$$



a b c d

FIGURE 3.16 Four image types and their corresponding histograms. (a) dark; (b) light; (c) low contrast; (d) high contrast. The horizontal axis of the histograms are values of r_k and the vertical axis are values of $p(r_k)$.

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Histogram Equalization

❖ Histogram equalization:

- To improve the contrast of an image
- To transform an image in such a way that the transformed image has a nearly uniform distribution of pixel values

❖ Transformation:

- Assume *r* has been normalized to the interval [0,L-1], with *r* = 0 representing black and *r* = L-1 representing white

$$s = T(r) \qquad 0 \le r \le L-1$$

- The transformation function satisfies the following conditions:
  - *T*(*r*) is single-valued and monotonically increasing in the interval
  - $0 \le T(r) \le L-1 \quad \text{for} \quad 0 \le r \le L-1$

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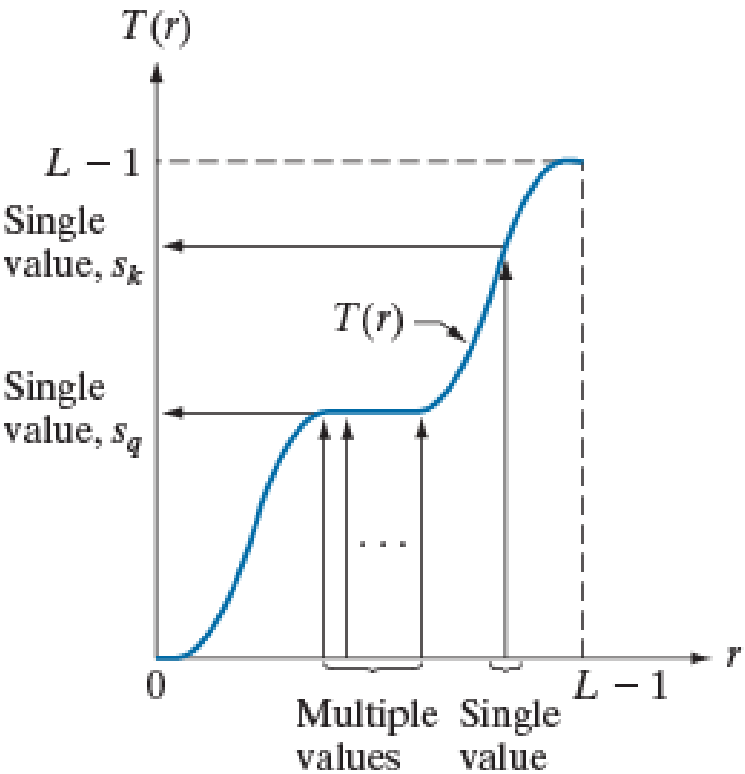
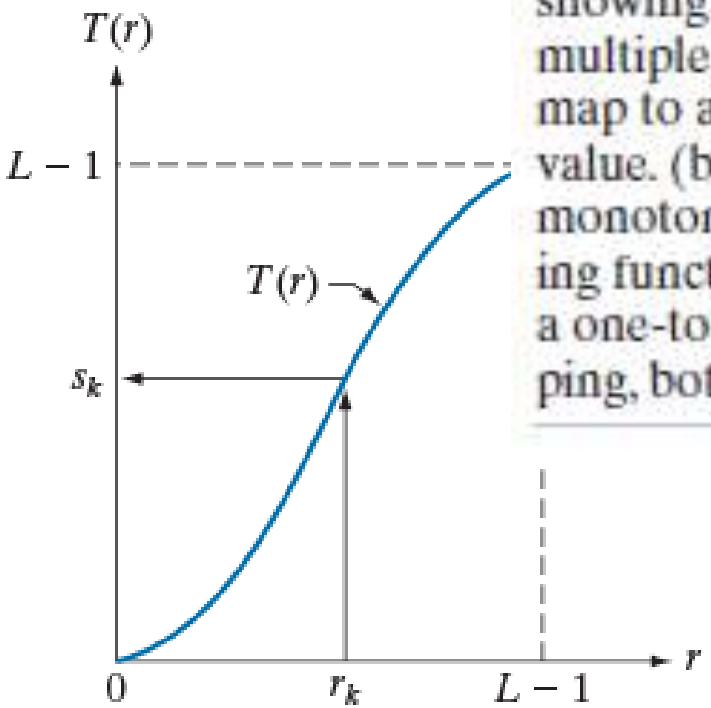
Histogram Equalization

❖ **For example:**





a b

FIGURE 3.17 (a) Monotonic increasing function, showing how multiple values can map to a single value. (b) Strictly monotonic increasing function. This is a one-to-one mapping, both ways.

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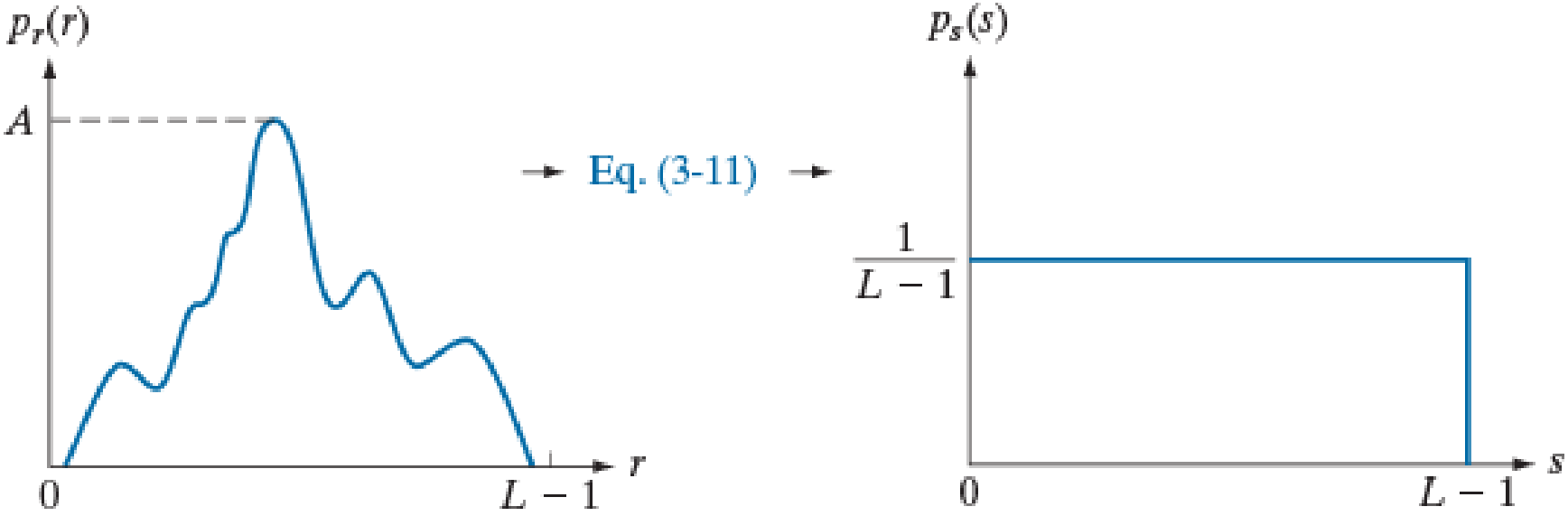
Histogram Equalization



a b

FIGURE 3.18 (a) An arbitrary PDF. (b) Result of applying Eq. (3-11) to the input PDF. The resulting PDF is always uniform, independently of the shape of the input.

為保護所參考來源之智慧財產權，此教材僅為上課時教學之用。

Histogram Equalization

- **Histogram equalization is based on a transformation of the probability density function of a random variable.**
- **Let $p_r(r)$ and $p_s(s)$ denote the probability density functions of random variable r and s, respectively.**

$$s = T(r) \qquad 0 \le r \le L-1$$

- **If $p_r(r)$ and $T(r)$ are known, then the probability density function $p_s(s)$ of the transformed variable s can be obtained**

$$p_s(s) = p_r(r)\left|\frac{dr}{ds}\right|$$

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Histogram Equalization

- **Define a transformation function where *w* is a dummy variable of integration and the right side of this equation is the cumulative distribution function of random variable *r*.**

$$s = T(r) = (L-1)\int_0^r p_r(w)dw$$

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Histogram Equalization

- **Given transformation function *T*(*r*),**

$$s = T(r) = (L-1)\int_0^r p_r(w)dw$$ *Leibniz's rule*

$$\frac{ds}{dr} = \frac{dT(r)}{dr} = (L-1)\frac{d}{dr}\left[\int_0^r p_r(w)dw\right] = (L-1)p_r(r)$$

$$\Rightarrow \quad p_s(s) = p_r(r)\left|\frac{dr}{ds}\right| = p_r(r)\left|\frac{1}{(L-1)p_r(r)}\right| = \frac{1}{(L-1)} \qquad 0 \le s \le L-1$$

$p_s(s)$ now is a uniform probability density function.

- **$T(r)$ depends on $p_r(r)$, but the resulting $p_s(s)$ always is uniform.**

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Another Interpretation

格列佛隧道（日語：ガリバートンネル）

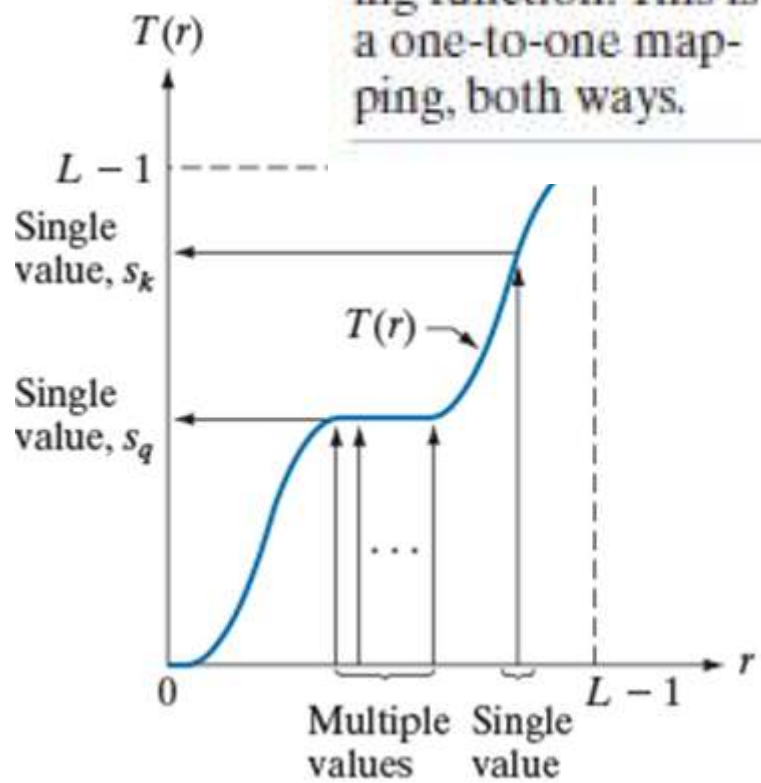
$\int_0^r P_r(w)dw = \int_0^s P_s(w)dw = S*1/(L-1)$

$S=(L-1)\int_0^r P_r(w)dw$

$S=T(r)=(L-1)\int_0^r P_r(w)dw$

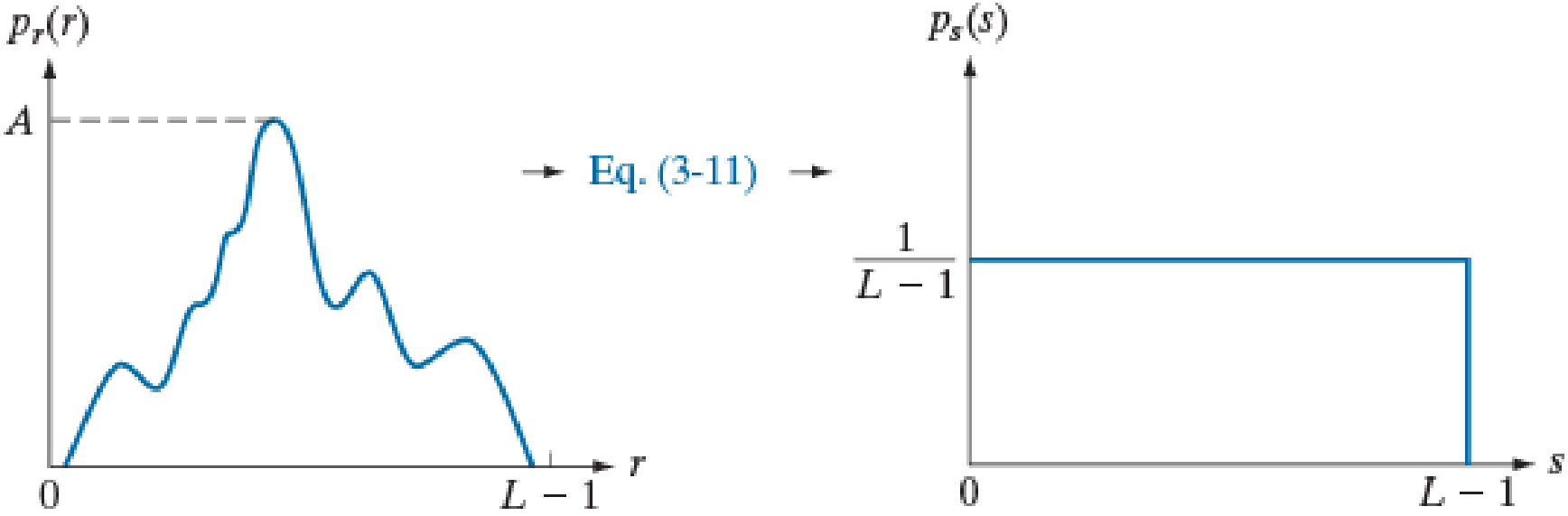
a b

FIGURE 3.17 (a) Monotonic increasing function, showing how multiple values can map to a single value. (b) Strictly monotonic increasing function. This is a one-to-one mapping, both ways.



為保護所參考來源之智慧財產權，此教材僅為上課時教學之用。

Histogram Equalization



a b

FIGURE 3.18 (a) An arbitrary PDF. (b) Result of applying Eq. (3-11) to the input PDF. The resulting PDF is always uniform, independently of the shape of the input.

為保護所參考來源之智慧財產權，此教材僅為上課時教學之用。

Histogram Equalization

$$s = T(r) = (L-1)\int_0^r p_r(w)dw$$

❖ In discrete version:

- The probability of occurrence of gray level r_k in an image is

$$p_r(r) = \frac{n_k}{n} \quad k = 0,1,2,\ldots,L-1$$

n : the total number of pixels in the image

n_k : the number of pixels that have gray level r_k

L : the total number of possible gray levels in the image

- The transformation function is

$$s_k = T(r_k) = (L-1)\sum_{j=0}^{k} p_r(r_j) = (L-1)\sum_{j=0}^{k}\frac{n_j}{n} \quad k = 0,1,2,\ldots,L-1$$

- Thus, an output image is obtained by mapping each pixel with level r_k in the input image into a corresponding pixel with level s_k.

為保護所參考來源之智慧財產權，此教材僅為上課時教學之用。

Histogram Equalization

$$s_k = T(r_k) = (L-1)\sum_{j=0}^{k}\frac{n_j}{n}$$

❖ For example:

TABLE 3.1 Intensity distribution and histogram values for a 3-bit, 64×64 digital image.

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

$$s_0 = T(r_0) = 7\sum_{j=0}^{0} p_r(r_j) = 7\,p_r(r_0) = 1.33 \approx 1$$

$$s_1 = T(r_1) = 7\sum_{j=0}^{1} p_r(r_j) = 7\,p_r(r_0) + 7\,p_r(r_1) = 3.08 \approx 3$$

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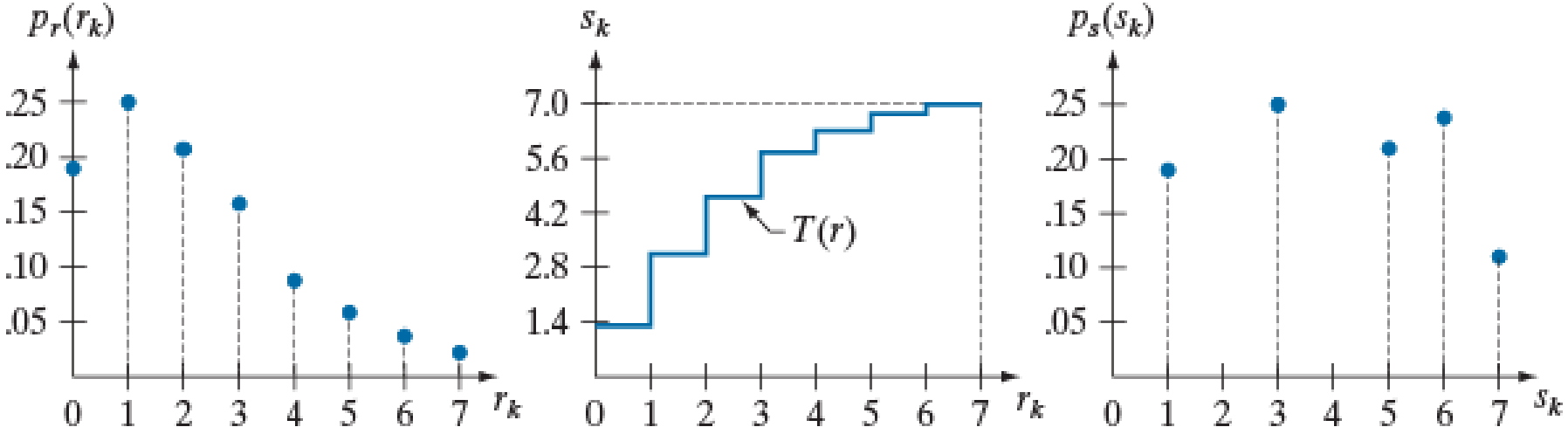
Histogram Equalization



a b c

FIGURE 3.19 Histogram equalization. (a) Original histogram. (b) Transformation function. (c) Equalized histogram.

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Histogram Equalization

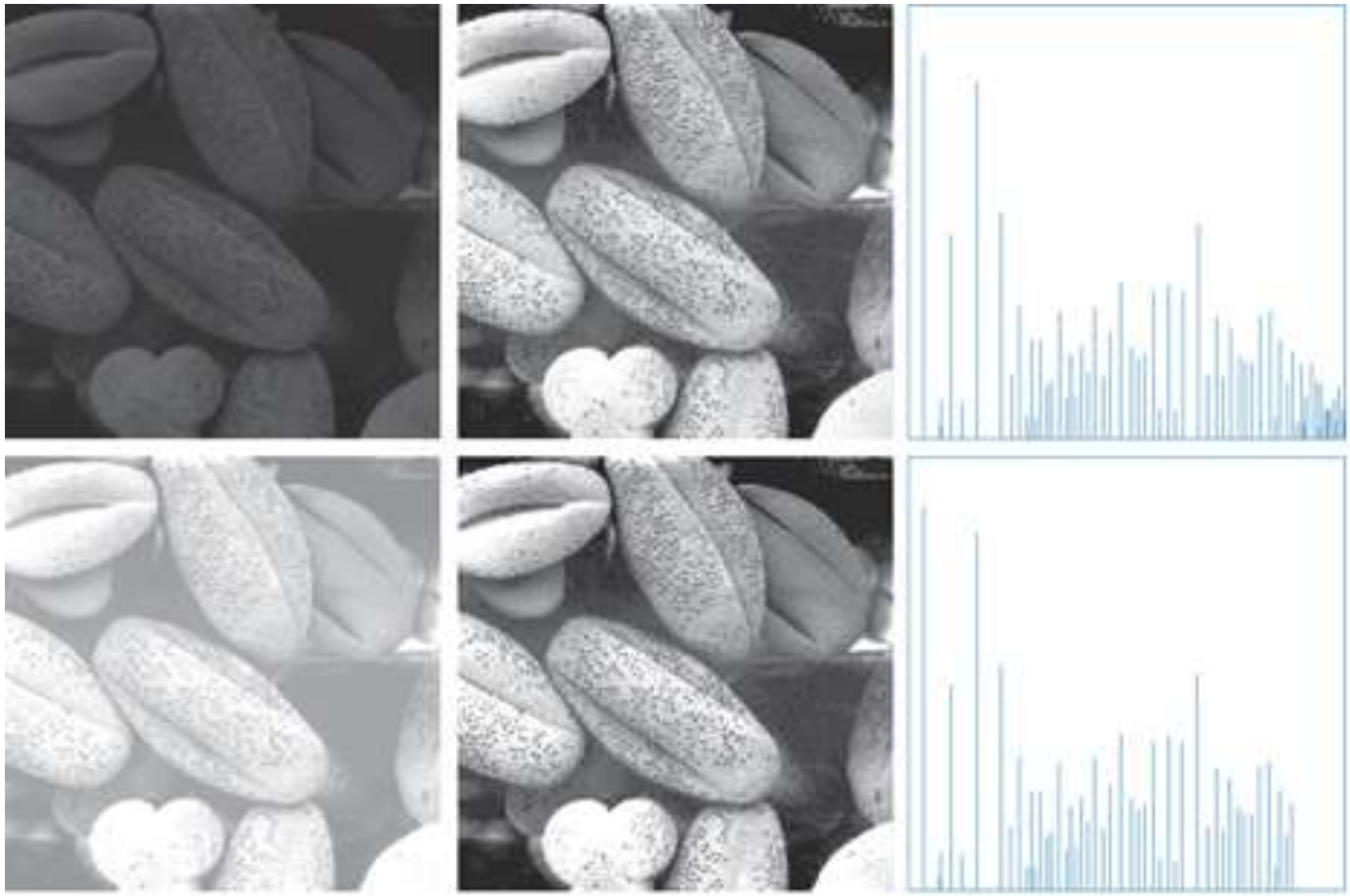

FIGURE 3.20 Left column: Images from Fig. 3.16. Center column: Corresponding histogram-equalized images. Right column: histograms of the images in the center column (compare with the histograms in Fig. 3.16).

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Histogram Equalization

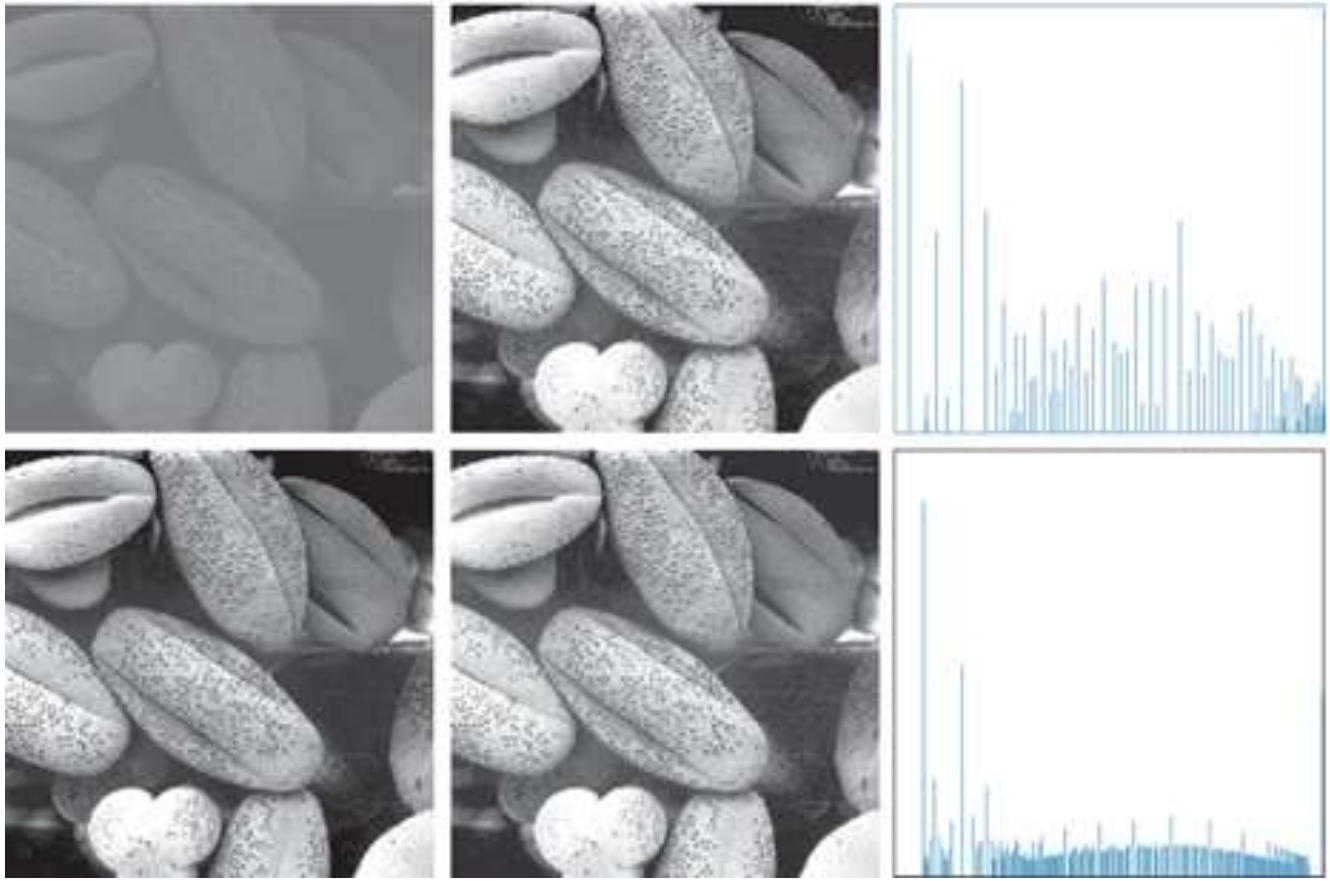

FIGURE 3.20 Left column: Images from Fig. 3.16. Center column: Corresponding histogram-equalized images. Right column: histograms of the images in the center column (compare with the histograms in Fig. 3.16).

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Histogram Equalization

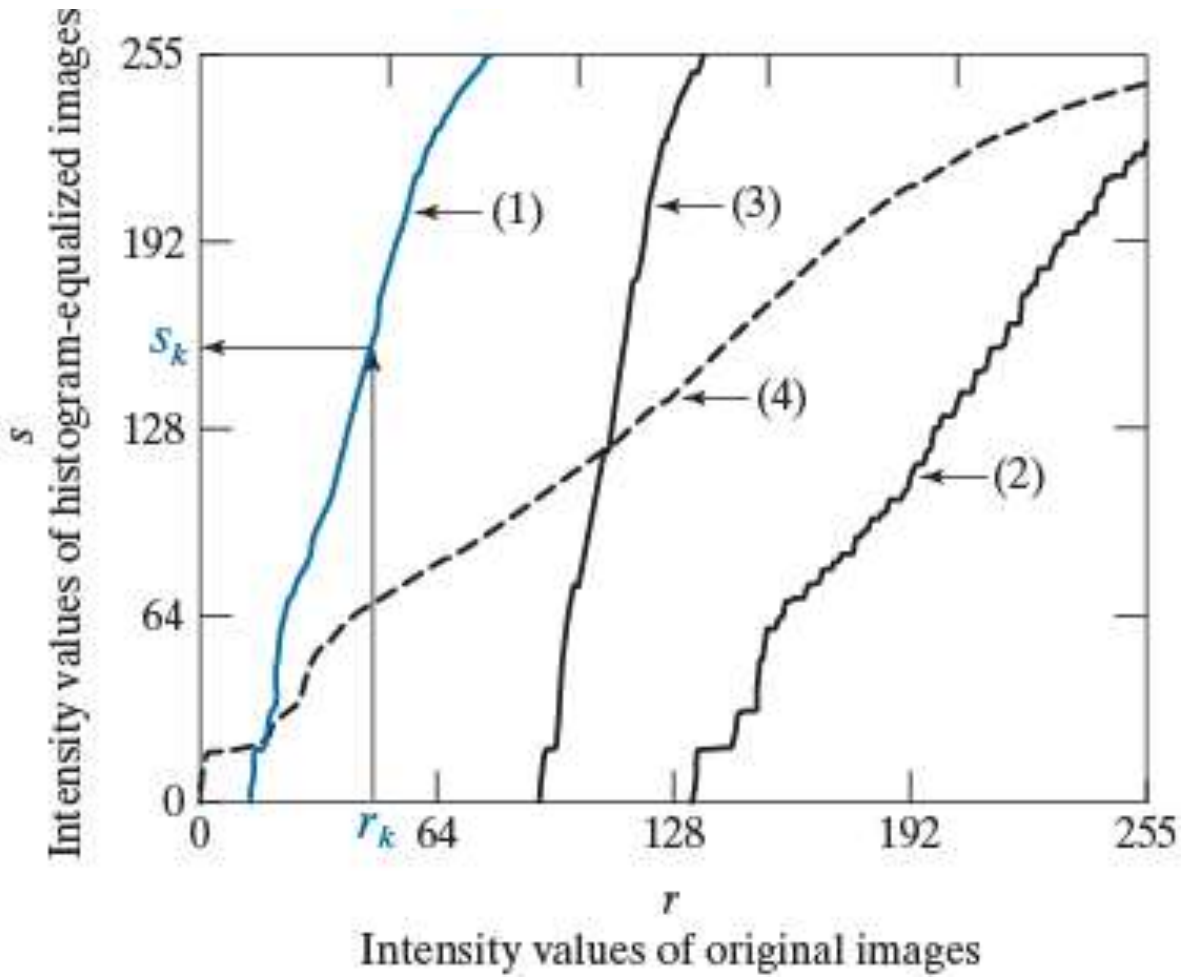

FIGURE 3.21 Transformation functions for histogram equalization. Transformations (1) through (4) were obtained using Eq. (3-15) and the histograms of the images on the left column of Fig. 3.20. Mapping of one intensity value r_k in image 1 to its corresponding value s_k is shown.

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Histogram Matching

- **Histogram matching is similar to histogram equalization, except that instead of trying to make the output image have a flat histogram, we would like it to have a histogram of a specified shape, say $p_z(z)$.**

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Histogram Matching

- **Obtain $p_r(r)$ from input image; and**

$$s = T(r) = (L-1)\int_0^r p_r(w)dw \qquad (1)$$

- **Obtain**

$$G(z) = (L-1)\int_0^z p_z(t)dt = s$$

- **Obtain z: $z=G^{-1}(s)$**

- **Equalizing input image by (1), pixel values are the *s* values**
  Inverse mapping $z=G^{-1}(s)$

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Histogram Matching

- **Obtain $p_r(r)$ from input image; and**

$$s_k = T(r_k) = (L-1)\sum_{j=0}^{k} p_r(r_j) = \frac{L-1}{MN}\sum_{j=0}^{k} n_j, k = 0,1,\ldots,L-1 \qquad (1)$$

- **Obtain**

$$G(z_q) = (L-1)\sum_{i=0}^{q} p_z(z_i) = s$$

- **Use equation above to map from *s* to *z***
  When more than one value of z_q satisfies the given s_k, choose the smallest value

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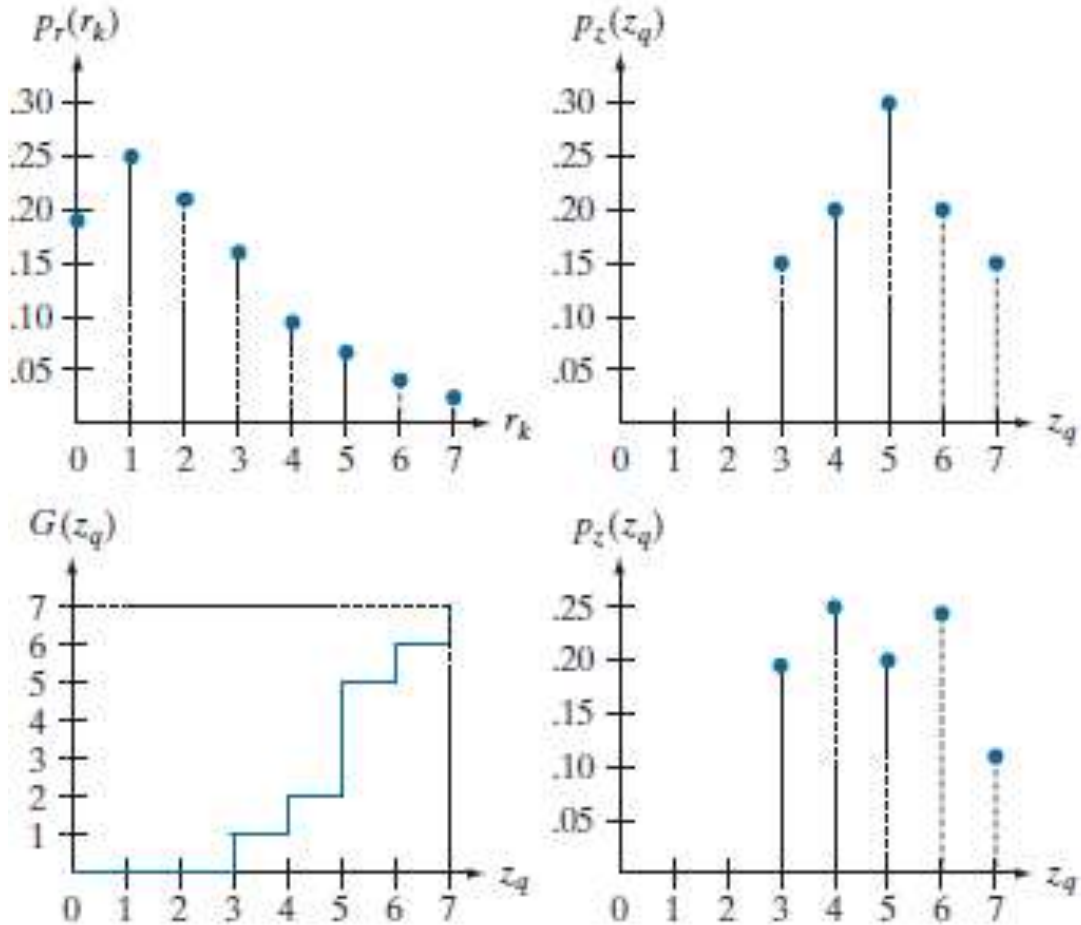
Histogram Matching

❖ **For example:**



a b
c d

FIGURE 3.22 (a) Histogram of a 3-bit image. (b) Specified histogram. (c) Transformation function obtained from the specified histogram. (d) Result of histogram specification. Compare the histograms in (b) and (d).

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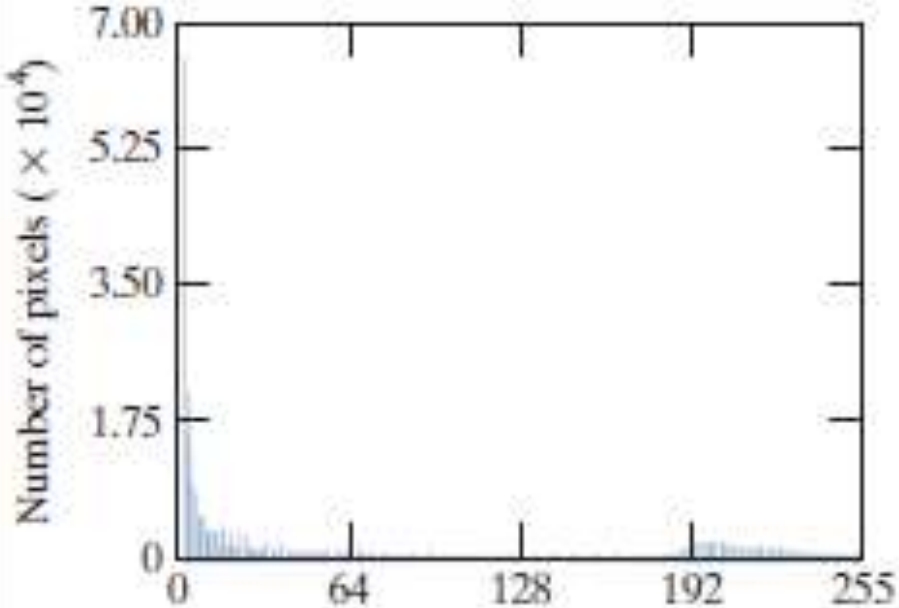
Histogram Matching

❖ **For example:**

a b

FIGURE 3.23
(a) An image, and
(b) its histogram.



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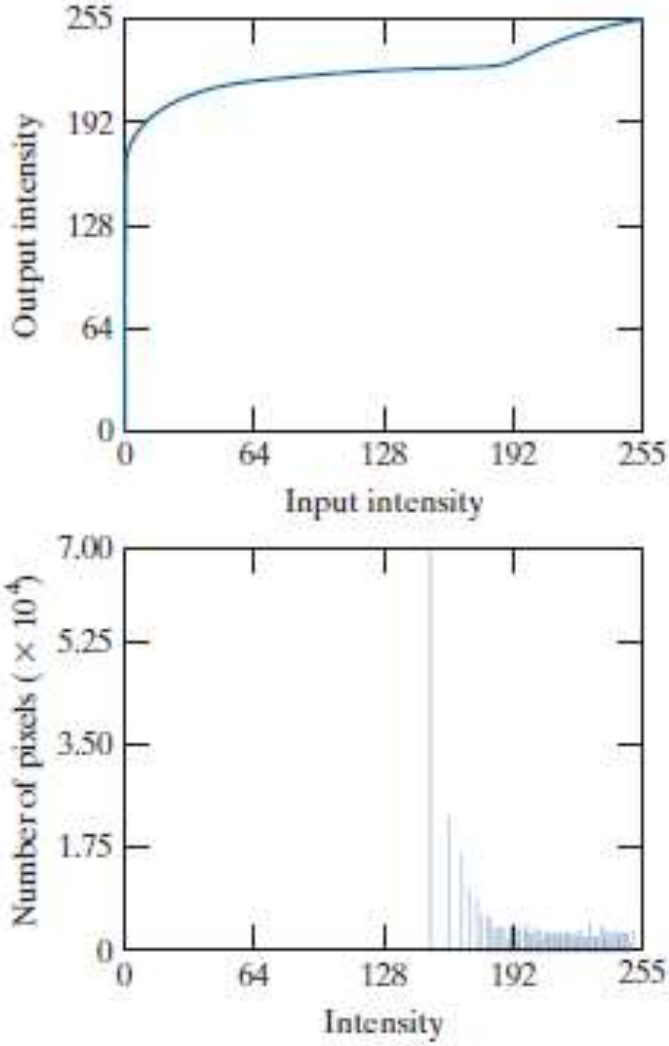
Histogram Matching



a b
c

FIGURE 3.24 (a) Histogram equalization transformation obtained using the histogram in Fig. 3.23(b). (b) Histogram equalized image. (c) Histogram of equalized image.

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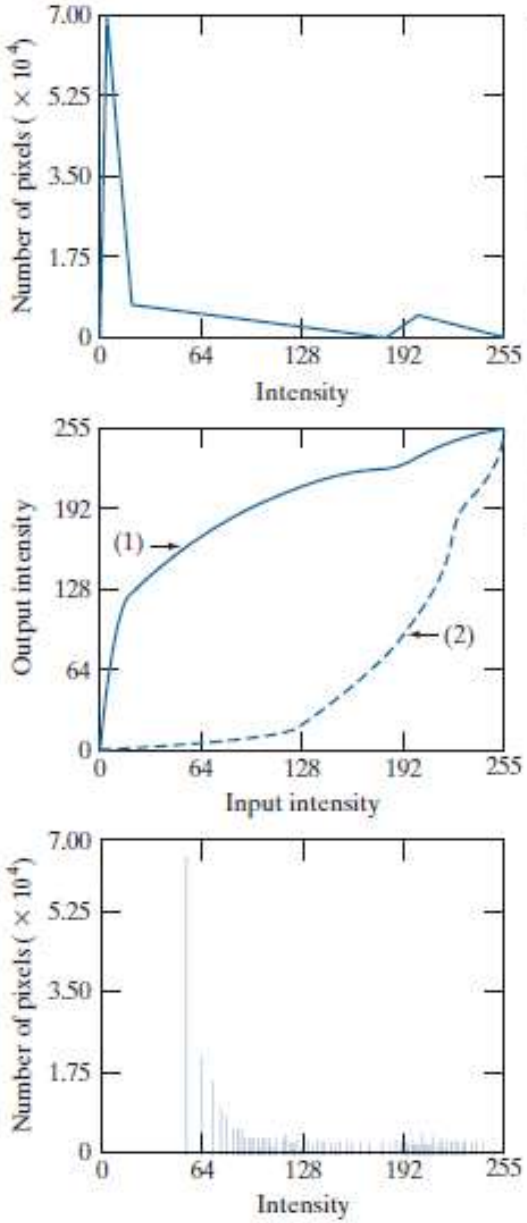
Histogram Matching



$$G(z_q) = (L-1)\sum_{i=0}^{q} p_z(z_i) \qquad (1)$$

$$z = G^{-1}(s) \qquad (2)$$

a c
b
d

FIGURE 3.25 Histogram specification. (a) Specified histogram. (b) Transformation $G(z_q)$, labeled (1), and $G^{-1}(s_k)$, labeled (2). (c) Result of histogram specification. (d) Histogram of image (c).

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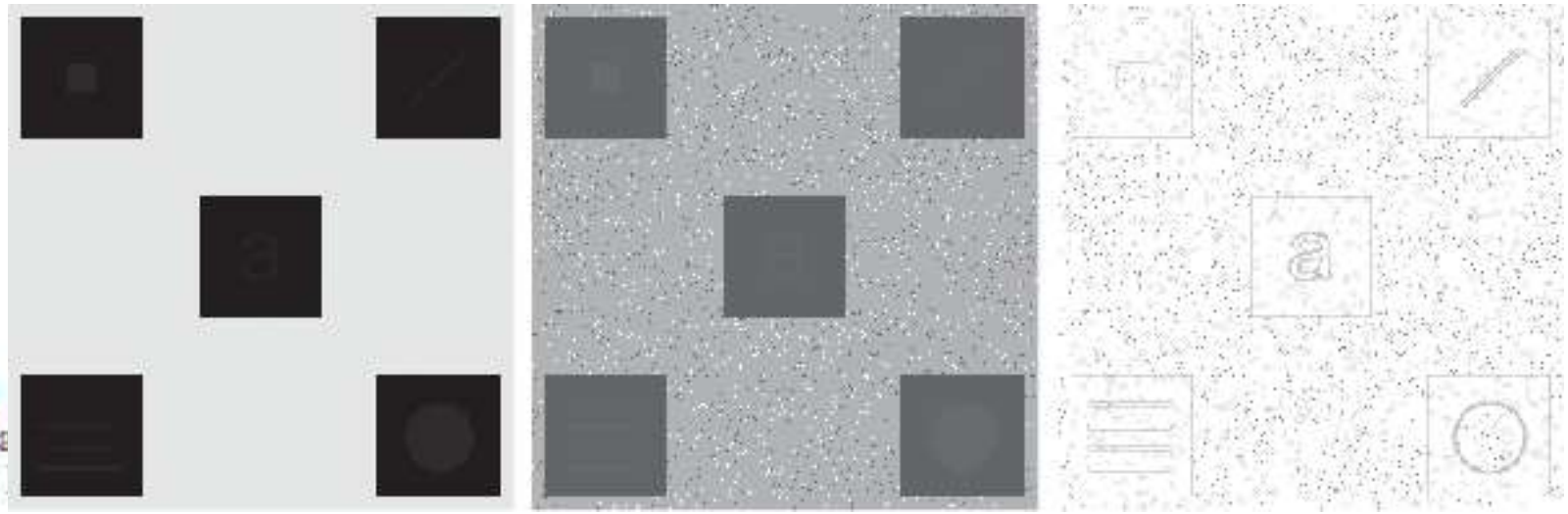
Local Histogram Processing

- The histogram processing methods discussed above are **global**, in the sense that pixels are modified by a transformation function **based on the gray-level content of an entire image.**
- However, there are cases in which it is necessary to enhance details over **small areas** in an image.



original global local

a b c

FIGURE 3.26 (a) Origina image. (b) of global histogram equalization. (c) Result of local histogram equalization.

Move pixel by pixel; update the central pixel using 3x3 histogram eq,

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Use of Histogram Statistics for Image Enhancement

- **Moments** can be determined directly from a histogram much faster than they can from the pixels directly.
- Let r denote a discrete random variable representing discrete gray-levels in the range $[0, L-1]$, and $p(r_i)$ denote the normalized histogram component corresponding to the ith value of r, then the nth **moment** of r about its mean is defined as

$$\mu_n(r) = \sum_{i=0}^{L-1} (r_i - m)^n p(r_i)$$

where m is the mean value of r

$$m = \sum_{i=0}^{L-1} r_i p(r_i)$$

- For example, the second moment (also the **variance** of r) is

$$\mu_2(r) = \sum_{i=0}^{L-1} (r_i - m)^2 p(r_i)$$

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Use of Histogram Statistics for Image Enhancement

- **Two uses of the mean and variance for enhancement purposes:**
  - The global mean and variance (global means for the entire image) are useful for adjusting overall contrast and intensity.
  - The mean and standard deviation for a local region are useful for correcting for large-scale changes in intensity and contrast

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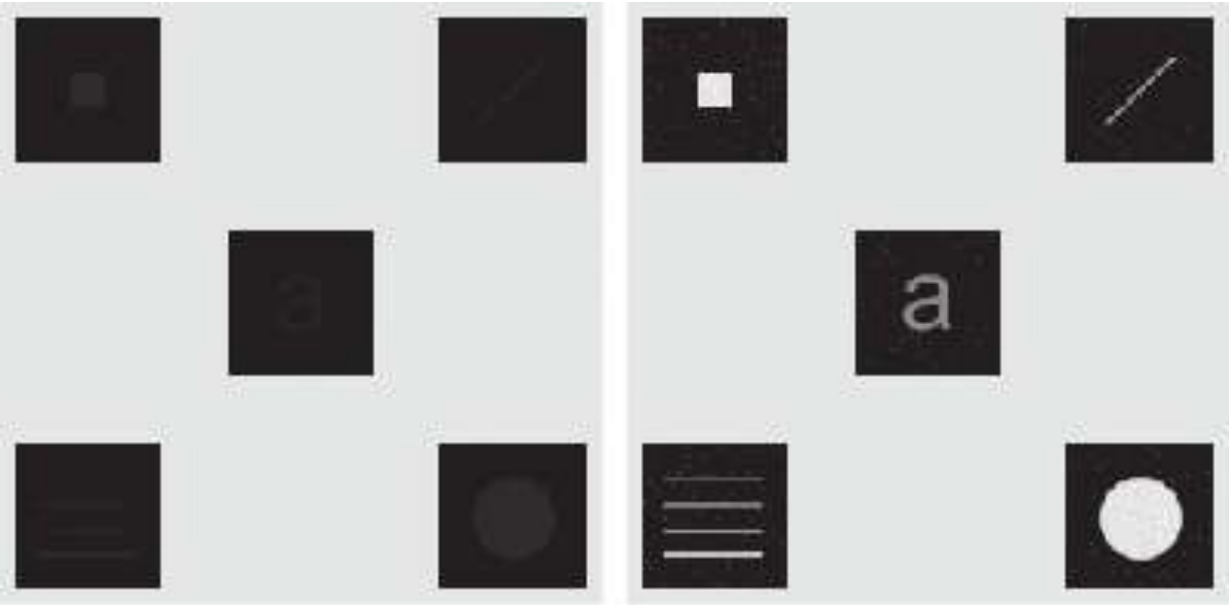
Use of Histogram Statistics for Image Enhancement



a b

FIGURE 3.27 (a) Original image. (b) Result of local enhancement based on local histogram statistics. Compare (b) with Fig. 3.26(c).

K_0=0.1, k_1=0 , k_2=0.1
C=22.8

$$g(x,y)=\begin{cases} Cf(x,y) \text{ if } \mathrm{ms}_{xy} \le k_0 m_G \text{ AND } \mathrm{k}_1\sigma_G \le \sigma_{S_{xy}} \le \mathrm{k}_2\sigma_G \\ f(x,y) \text{ otherwise} \end{cases}$$

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Spatial Filtering

- In spatial filtering (vs. frequency domain filtering), the output image is computed directly by simple calculations on the pixels of the input image.
- Spatial filtering can be either linear or non-linear.
- For each output pixel, some neighborhood of input pixels is used in the computation.
- In general, linear filtering of an image *f* of size *MXN* with a filter mask of size *mxn* is given by

$$g(x,y)=\sum_{s=-a}^{a}\sum_{t=-b}^{b}w(s,t)f(x+s,y+t)$$

where $a=(m-1)/2$ and $b=(n-1)/2$

- convolution

$$g(x,y)=\sum_{s=-a}^{a}\sum_{t=-b}^{b}w(s,t)f(x-s,y-t)$$

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Spatial Filtering

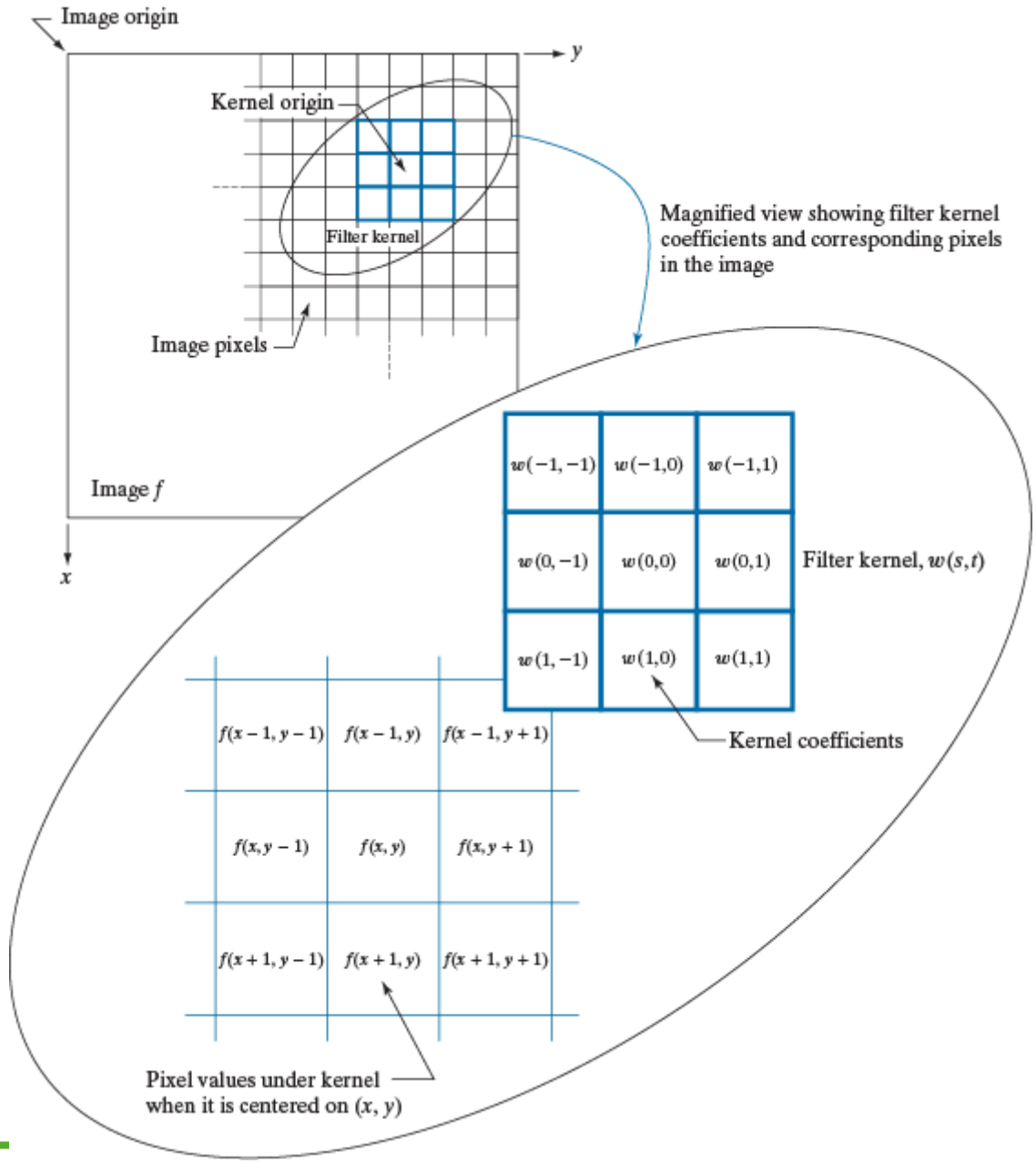

FIGURE 3.28 The mechanics of linear spatial filtering using a 3×3 kernel. The pixels are shown as squares to simplify the graphics. Note that the origin of the image is at the top left, but the origin of the kernel is at its center. Placing the origin at the center of spatially symmetric kernels simplifies writing expressions for linear filtering.

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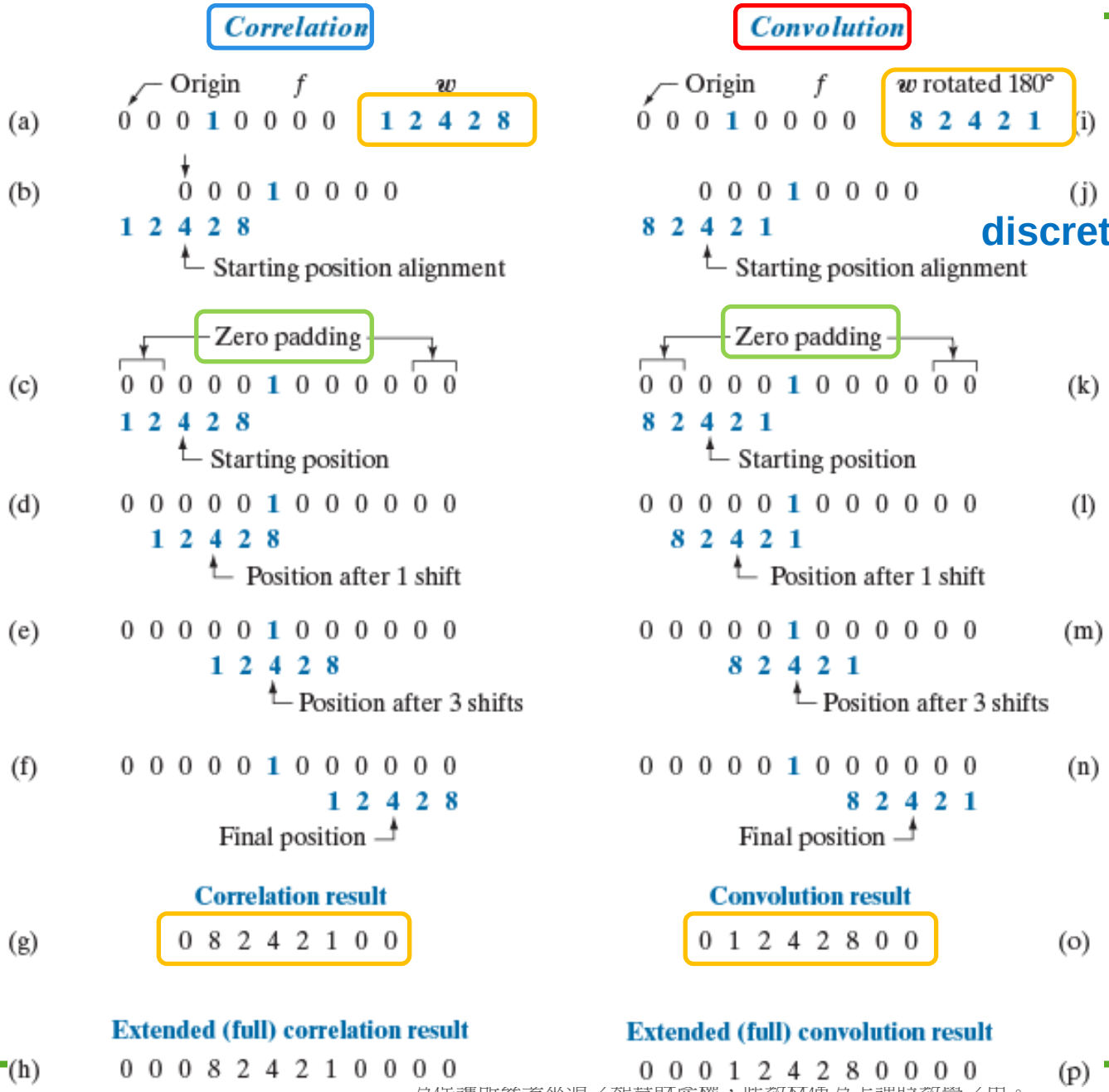
Correlation and Convolution



discrete unit impulse

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Correlation and Convolution

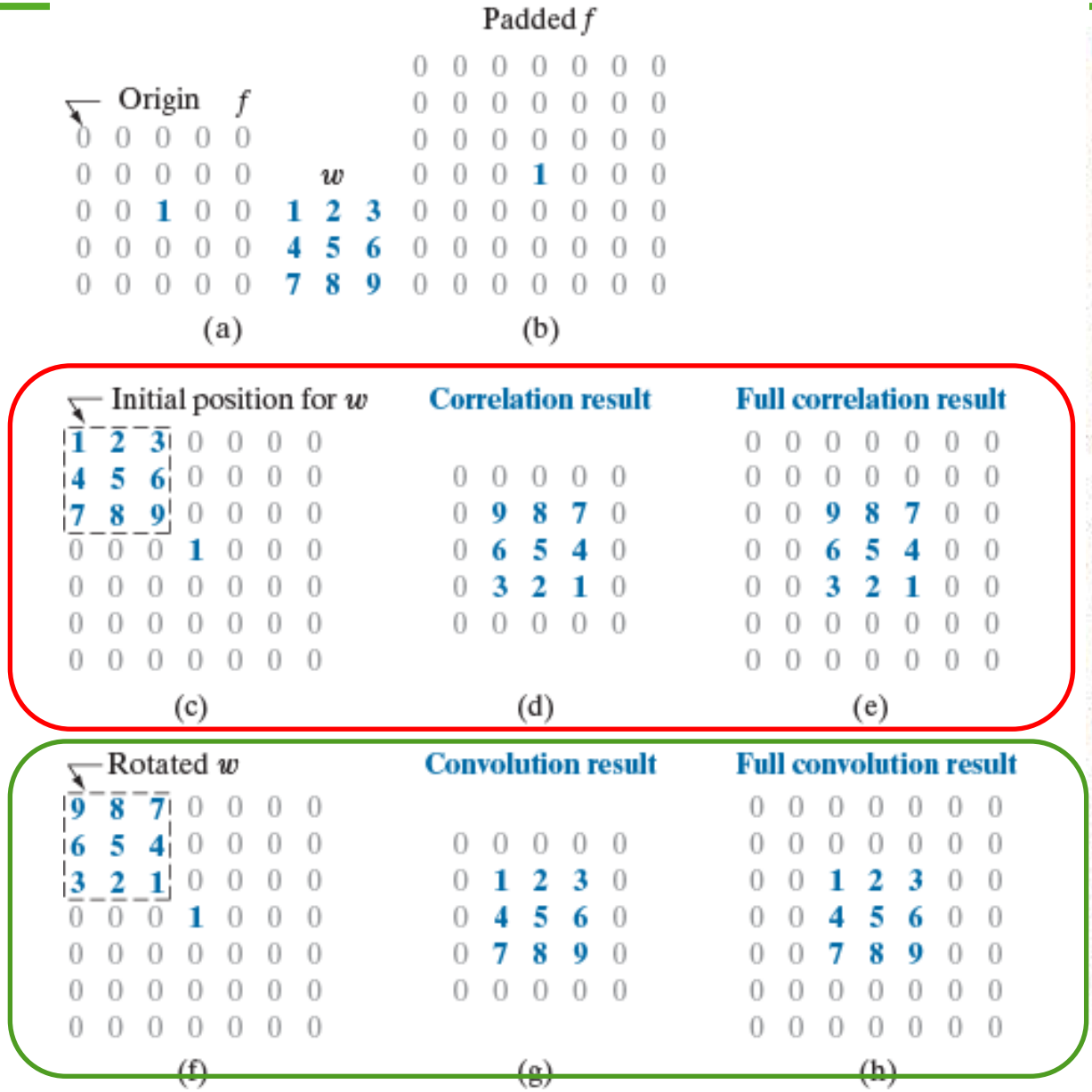

FIGURE 3.30 Correlation (middle row) and convolution (last row) of a 2-D kernel with an image consisting of a discrete unit impulse. The 0's are shown in gray to simplify visual analysis. Note that correlation and convolution are functions of x and y. As these variable change, they *displace* one function with respect to the other. See the discussion of Eqs. (3-36) and (3-37) regarding full correlation and convolution.

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Smoothing Linear Filters

❖ Smoothing linear filters

- Averaging filters
  - (a) Box filter
  - (b) Gaussian kernel filter

$$\frac{1}{9} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \qquad \frac{1}{4.8976} \times \begin{bmatrix} 0.3679 & 0.6065 & 0.3679 \\ 0.6065 & 1.0000 & 0.6065 \\ 0.3679 & 0.6065 & 0.3679 \end{bmatrix}$$

a b

FIGURE 3.31 Examples of smoothing kernels: (a) is a *box* kernel; (b) is a *Gaussian* kernel.

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Smoothing Linear Filters

- **The general implementation for filtering an *MXN* image with a weighted averaging filter of size *mxn* is given by**

$$g(x,y)=\frac{\sum_{s=-a}^{a}\sum_{t=-b}^{b}w(s,t)f(x+s,y+t)}{\sum_{s=-a}^{a}\sum_{t=-b}^{b}w(s,t)}$$

where *a*=(*m*-1)/2 and *b*=(*n*-1)/2

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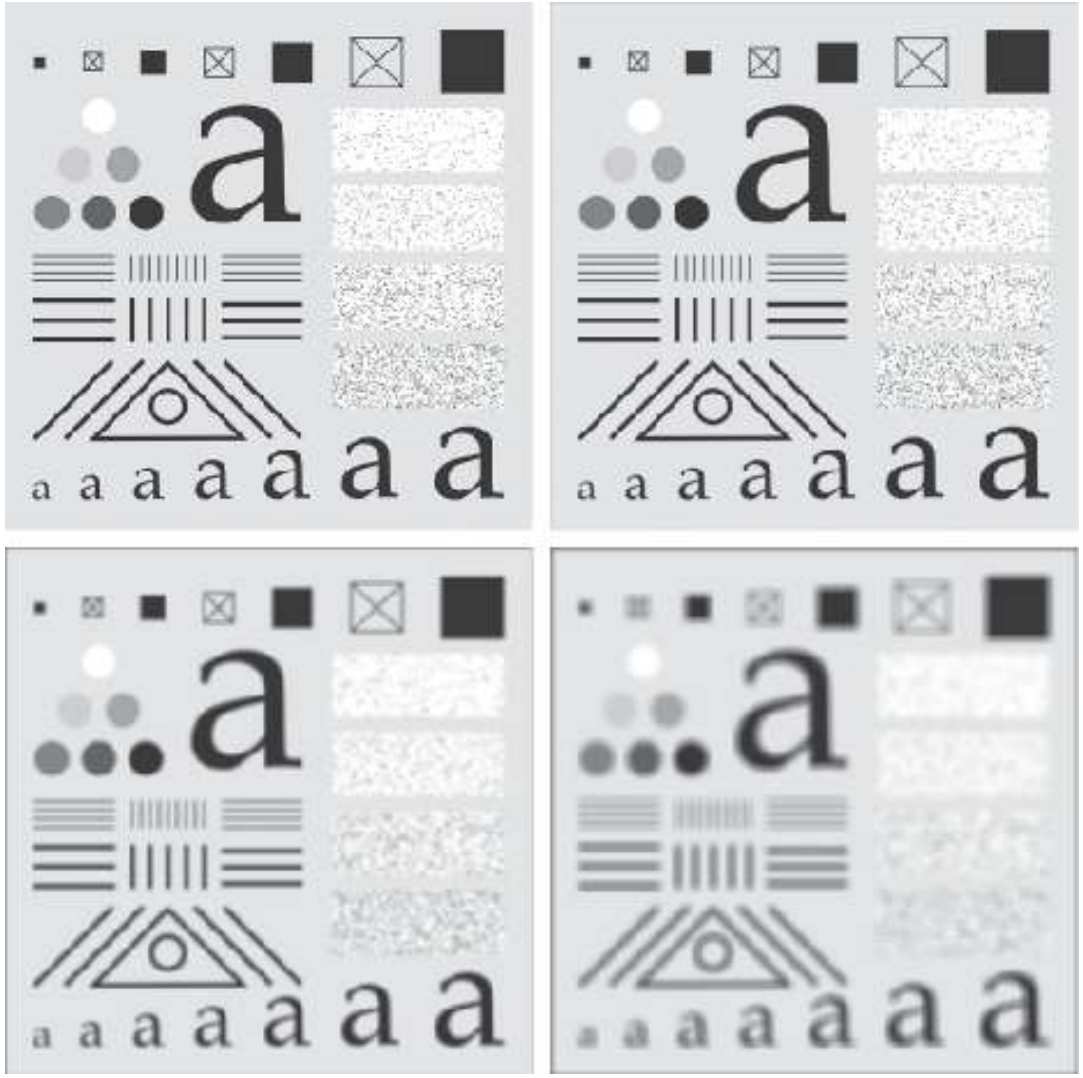
Smoothing Linear Filters

masks of various sizes



a b
c d

FIGURE 3.33 (a) Test pattern of size 1024×1024 pixels. (b)-(d) Results of lowpass filtering with box kernels of sizes 3×3, 11×11, and 21×21, respectively.

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Smoothing Linear Filters
masks of various sizes

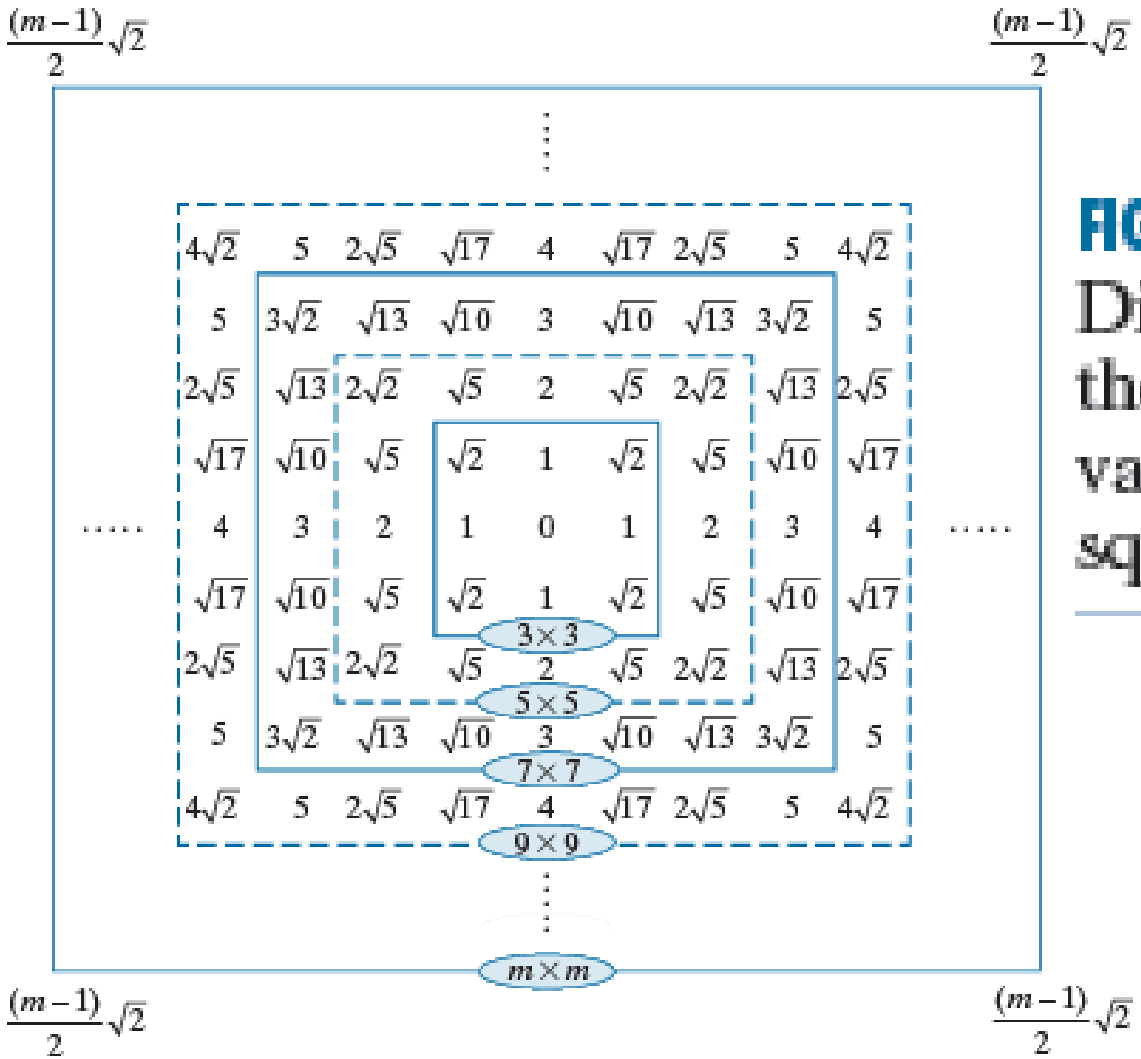

FIGURE 3.34 Distances from the center for various sizes of square kernels.

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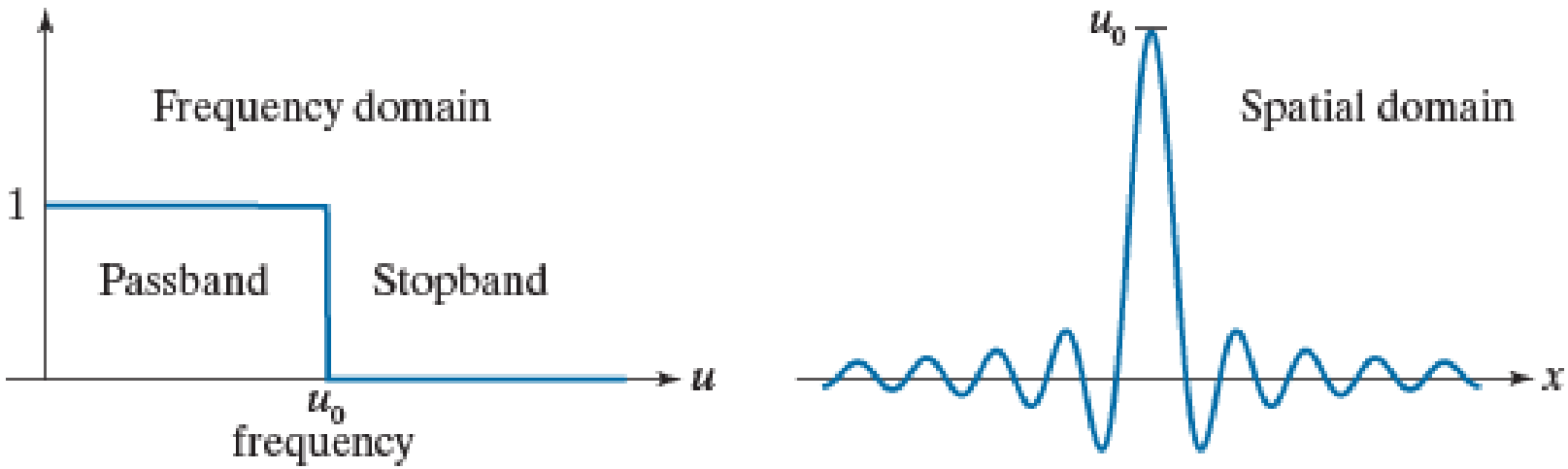
Smoothing Linear Filters
Filtering (more in Ch4)



a b

FIGURE 3.32 (a) Ideal 1-D low-pass filter transfer function in the frequency domain. (b) Corresponding filter kernel in the spatial domain.

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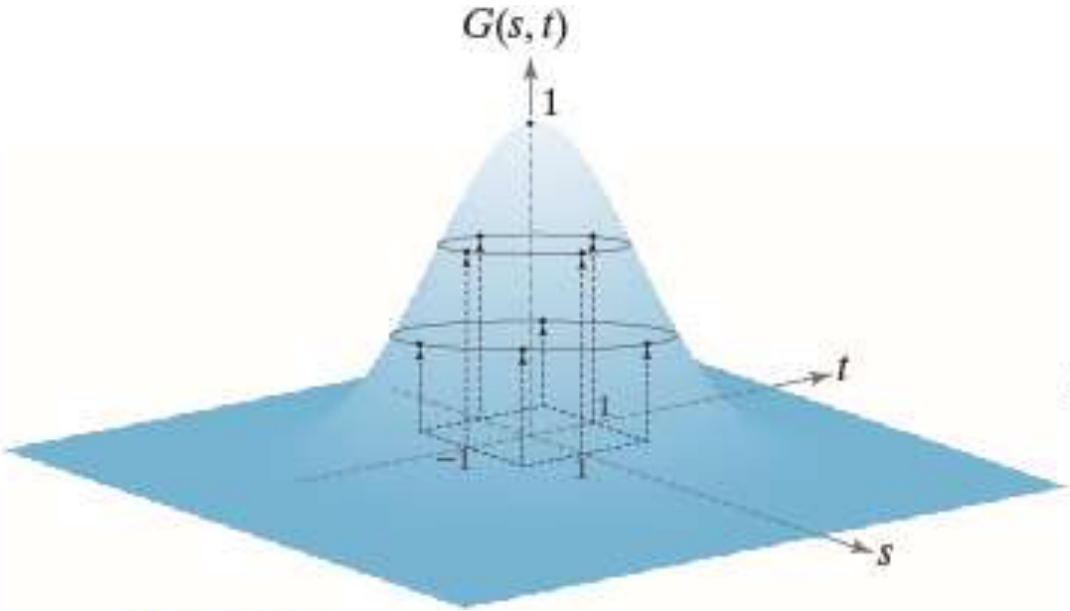
Smoothing Linear Filters

Gaussian kernel



$$\frac{1}{4.8976} \times \begin{bmatrix} 0.3679 & 0.6065 & 0.3679 \\ 0.6065 & 1.0000 & 0.6065 \\ 0.3679 & 0.6065 & 0.3679 \end{bmatrix}$$

FIGURE 3.35 (a) Sampling a Gaussian function to obtain a discrete Gaussian kernel. The values shown are for $K = 1$ and $\sigma = 1$. (b) Resulting 3×3 kernel [this is the same as Fig. 3.31(b)].

$$\omega(s,t) = Ke^{-\frac{s^2+t^2}{2\sigma^2}}$$

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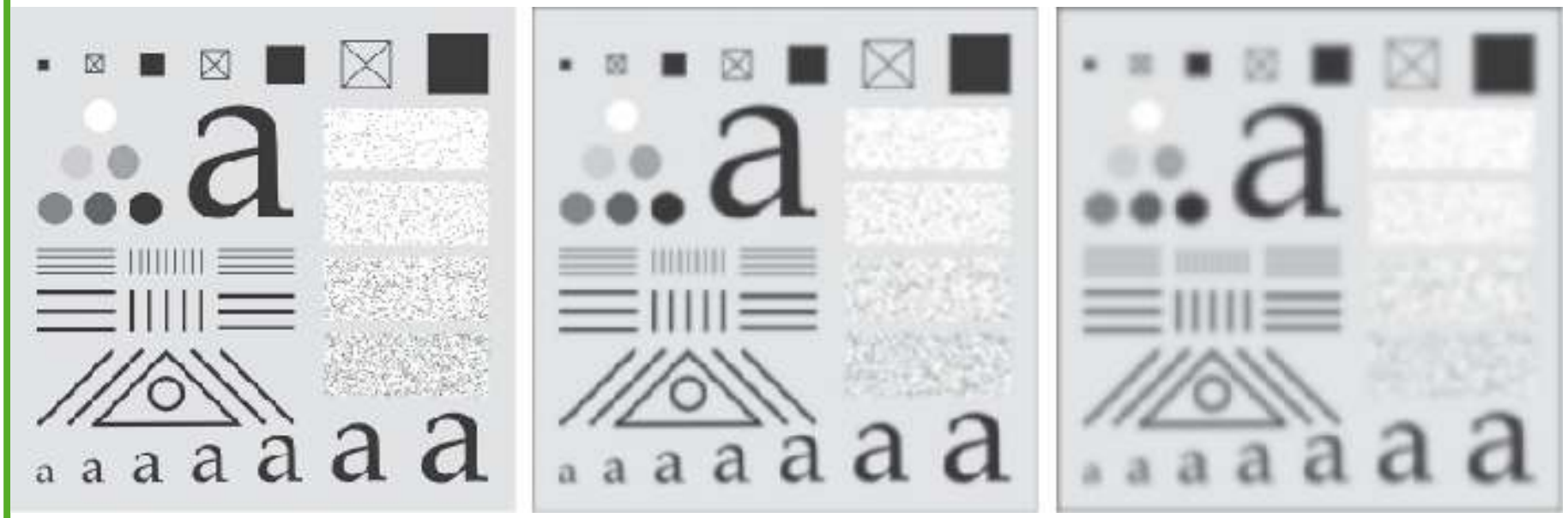
Smoothing Linear Filters

Gaussian kernel

$$\omega(s,t) = Ke^{-\frac{s^2+t^2}{2\sigma^2}}$$



a b c

FIGURE 3.36 (a)A test pattern of size 1024×1024. (b) Result of lowpass filtering the pattern with a Gaussian kernel of size 21×21, with standard deviations $\sigma = 3.5$. (c) Result of using a kernel of size 43×43, with $\sigma = 7$. This result is comparable to Fig. 3.33(d). We used $K = 1$ in all cases.

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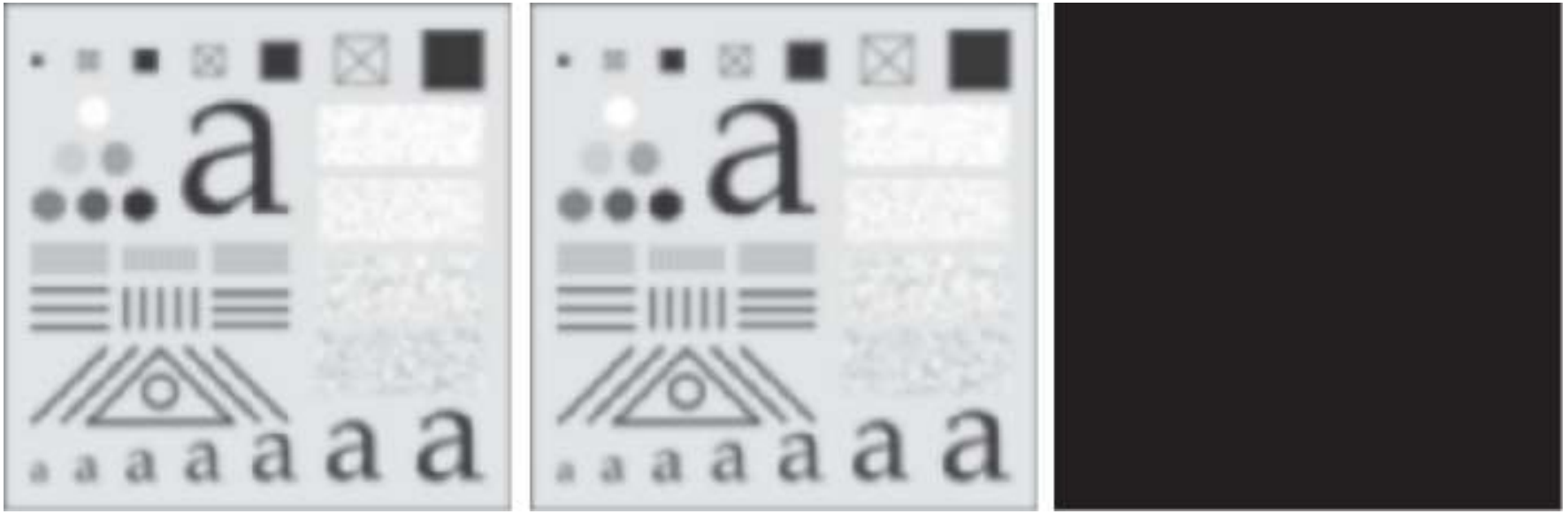
Smoothing Linear Filters
Gaussian kernel

$$\omega(s,t) = Ke^{-\frac{s^2+t^2}{2\sigma^2}}$$



a b c

FIGURE 3.37 (a) Result of filtering Fig. 3.36(a) using a Gaussian kernels of size 43×43, with $\sigma = 7$. (b) Result of using a kernel of 85×85, with the same value of σ. (c) Difference image.

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Smoothing Linear Filters

Region extraction

a b c

FIGURE 3.41 (a) A 2566 × 2758 Hubble Telescope image of the *Hickson Compact Group*. (b) Result of lowpass filtering with a Gaussian kernel. (c) Result of thresholding the filtered image (intensities were scaled to the range [0, 1]). The Hickson Compact Group contains dwarf galaxies that have come together, setting off thousands of new star clusters. (Original image courtesy of NASA.)

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Smoothing Linear Filters

Shading correction

a b c

FIGURE 3.42 (a) Image shaded by a shading pattern oriented in the −45° direction. (b) Estimate of the shading patterns obtained using lowpass filtering. (c) Result of dividing (a) by (b). (See Section 9.8 for a morphological approach to shading correction).

512x512 Gaussian kernel, K=1, σ=128 (same size of the squares, 128x128)

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Spatial Filtering

- **Nonlinear spatial filtering usually uses a neighborhood too, but some other mathematical operations are used. These can include conditional operations (if …, then…), statistical (sorting pixel values in the neighborhood), etc.**
- **Because the neighborhood includes pixels on all sides of the center pixel, some special procedure must be used along the top, bottom, left and right sides of the image so that the processing does not try to use pixels that do not exist.**

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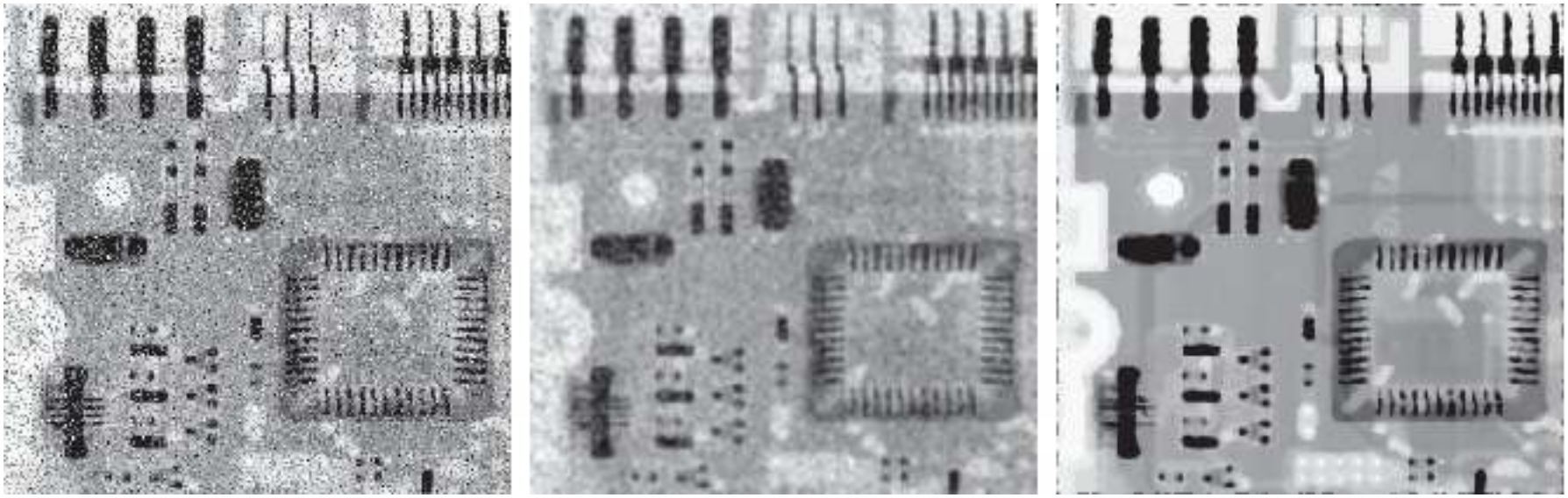
Order-Statistic Filters

❖ Order-statistic filters

- Median filter: to reduce impulse noise (salt-and-pepper noise)



a b c

FIGURE 3.43 (a) X-ray image of a circuit board, corrupted by salt-and-pepper noise. (b) Noise reduction using a 19×19 Gaussian lowpass filter kernel with $\sigma = 3$. (c) Noise reduction using a 7×7 median filter. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)

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First/Second-Order Derivative

- **The first-order derivative**
  - Areas of constant intensity: zero
  - Onset of an intensity step or ramp: nonzero
  - Along ramps: nonzero

- **The second-order**
  - Areas of constant intensity: zero
  - Onset and end of an intensity step or ramp: nonzero
  - Along ramps: zero

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Sharpening Spatial Filters

- **Sharpening filters are based on computing spatial derivatives of an image.**
- **The first-order derivative of a one-dimensional function *f*(*x*) is**

$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

- **The second-order derivative of a one-dimensional function *f*(*x*) is**

$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$

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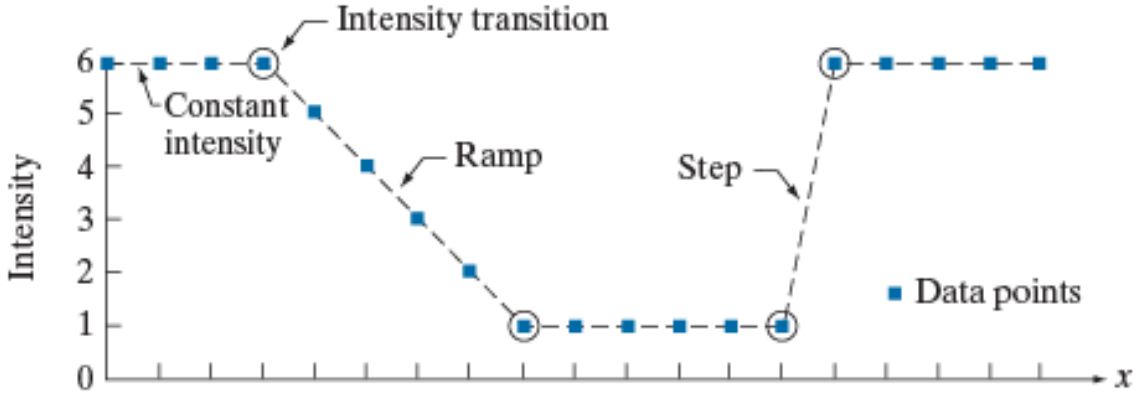
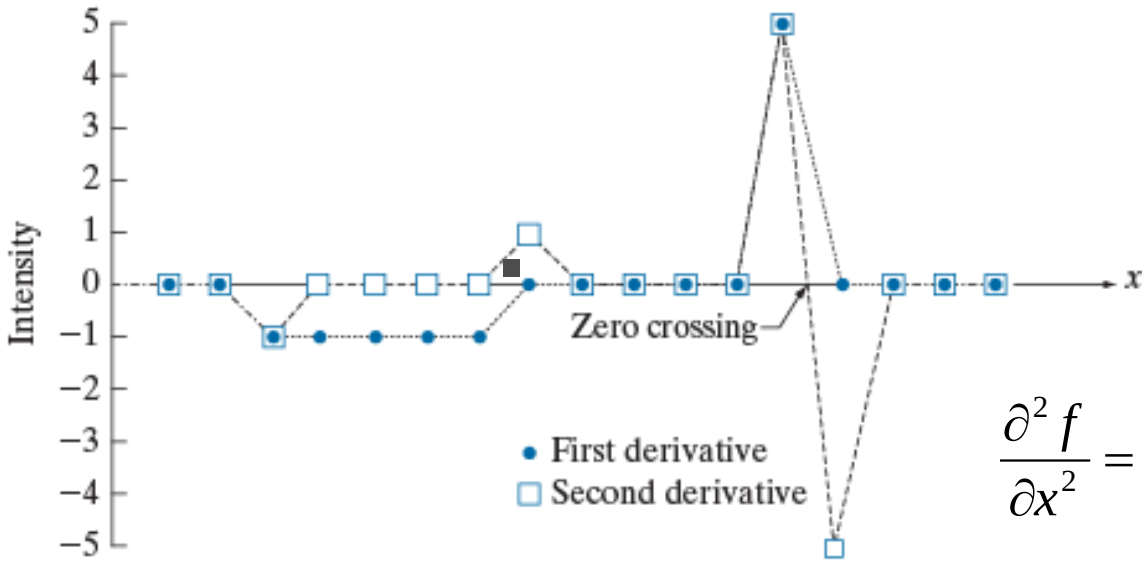
Sharpening Spatial Filters



Values of scan line	6	6	6	6	5	4	3	2	1	1	1	1	1	1	6	6	6	6	6
1st derivative		0	0	−1	−1	−1	−1	−1	0	0	0	0	0	5	0	0	0	0	
2nd derivative		0	0	−1	0	0	0	0	1	0	0	0	0	5	−5	0	0	0	



a
b
c

FIGURE 3.44 (a) A section of a horizontal scan line from an image, showing ramp and step edges, as well as constant segments. (b) Values of the scan line and its derivatives. (c) Plot of the derivatives, showing a zero crossing. In (a) and (c) points were joined by dashed lines as a visual aid.

$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$

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Use of Second Derivatives
The Laplacian

❖ Development of the Laplacian method

- The two dimensional Laplacian operator for continuous functions:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

- The Laplacian is a linear operator.

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

$$\nabla^2 f = [f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1)] - 4f(x)$$

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Use of Second Derivatives
The Laplacian

0	1	0
1	−4	1
0	1	0

1	1	1
1	−8	1
1	1	1

0	−1	0
−1	4	−1
0	−1	0

−1	−1	−1
−1	8	−1
−1	−1	−1

a b c d

FIGURE 3.45 (a) Laplacian kernel used to implement Eq. (3-53). (b) Kernel used to implement an extension of this equation that includes the diagonal terms. (c) and (d) Two other Laplacian kernels.

$$\nabla^2 f = [f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1)] - 4f(x)$$

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Use of Second Derivatives
The Laplacian

- **To sharpen an image, the Laplacian of the image is subtracted from the original image.**

$$g(x,y)=\begin{cases} f(x,y)-\nabla^2 f & \text{if the center coefficient of the Laplacian mask is negative.} \\ f(x,y)+\nabla^2 f & \text{if the center coefficient of the Laplacian mask is positive.} \end{cases}$$

$$g(x,y)=f(x,y)+c\left[\nabla^2 f(x,y)\right]$$

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Use of Second Derivatives
The Laplacian

Negative values show as black by the display

a b
c d

FIGURE 3.46 (a) Blurred image of the North Pole of the moon. (b) Laplacian image obtained using the kernel in Fig. 3.45(a). (c) Image sharpened using Eq. (3-54) with $c = -1$. (d) Image sharpened using the same procedure, but with the kernel in Fig. 3.45(b). (Original image courtesy of NASA.)

FIGURE 3.47 The Laplacian image from Fig. 3.46(b), scaled to the full [0, 255] range of intensity values. Black pixels correspond to the most negative value in the unscaled Laplacian image, grays are intermediate values, and white pixels corresponds to the highest positive value.

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Unsharp Masking and Highboost Filtering

- **Blur the original image** $\overline{f}(x,y)$
- **Obtain mask: subtract the blurred image from the original**
- **Add the mask to the original**
  - k=1: unsharp masking
  - k>1: highboost filtering

$$g_{mask}(x,y) = f(x,y) - \overline{f}(x,y)$$

$$g(x,y) = f(x,y) + k \cdot g_{mask}(x,y)$$

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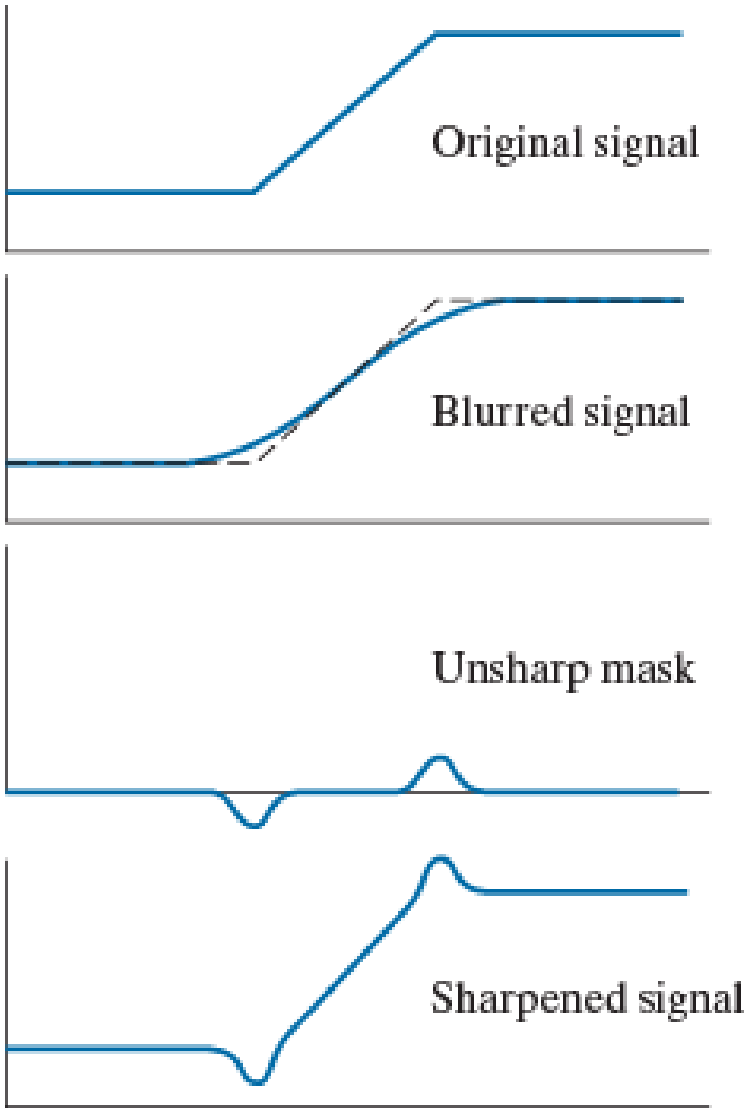
Unsharp Masking and Highboost Filtering



a
b
c
d

FIGURE 3.48 1-D illustration of the mechanics of unsharp masking. (a) Original signal. (b) Blurred signal with original shown dashed for reference. (c) Unsharp mask. (d) Sharpened signal, obtained by adding (c) to (a).

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Unsharp Masking and Highboost Filtering

a b c
d e

K=1 K=4.5

FIGURE 3.49 (a) Original image of size 600×259 pixels. (b) Image blurred using a 31×31 Gaussian lowpass filter with $\sigma = 5$. (c) Mask. (d) Result of unsharp masking using Eq. (3-56) with $k = 1$. (e) Result of highboost filtering with $k = 4.5$.

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First Derivatives
The Gradient

❖ Development of the Gradient method

- The gradient of function f at coordinates (x,y) is defined as the two-dimensional column vector:

$$\nabla \mathbf{f} = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \dfrac{\partial f}{\partial x} \\ \dfrac{\partial f}{\partial y} \end{bmatrix}$$

- The magnitude of this vector is given by

$$M(x,y) = \mathrm{mag}(\nabla \mathbf{f}) = \left[G_x^2 + G_y^2\right]^{\frac{1}{2}} = \left[\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2\right]^{\frac{1}{2}}$$

$$M(x,y) \approx |G_x| + |G_y|$$

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First Derivatives
The Gradient

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

Roberts cross-gradient operators

−1	0
0	1

0	−1
1	0

Sobel operators

−1	−2	−1
0	0	0
1	2	1

−1	0	1
−2	0	2
−1	0	1

a
b c
d e

FIGURE 3.50 (a) A 3 × 3 region of an image, where the *z*s are intensity values. (b)–(c) Roberts cross-gradient operators. (d)–(e) Sobel operators. All the kernel coefficients sum to zero, as expected of a derivative operator.

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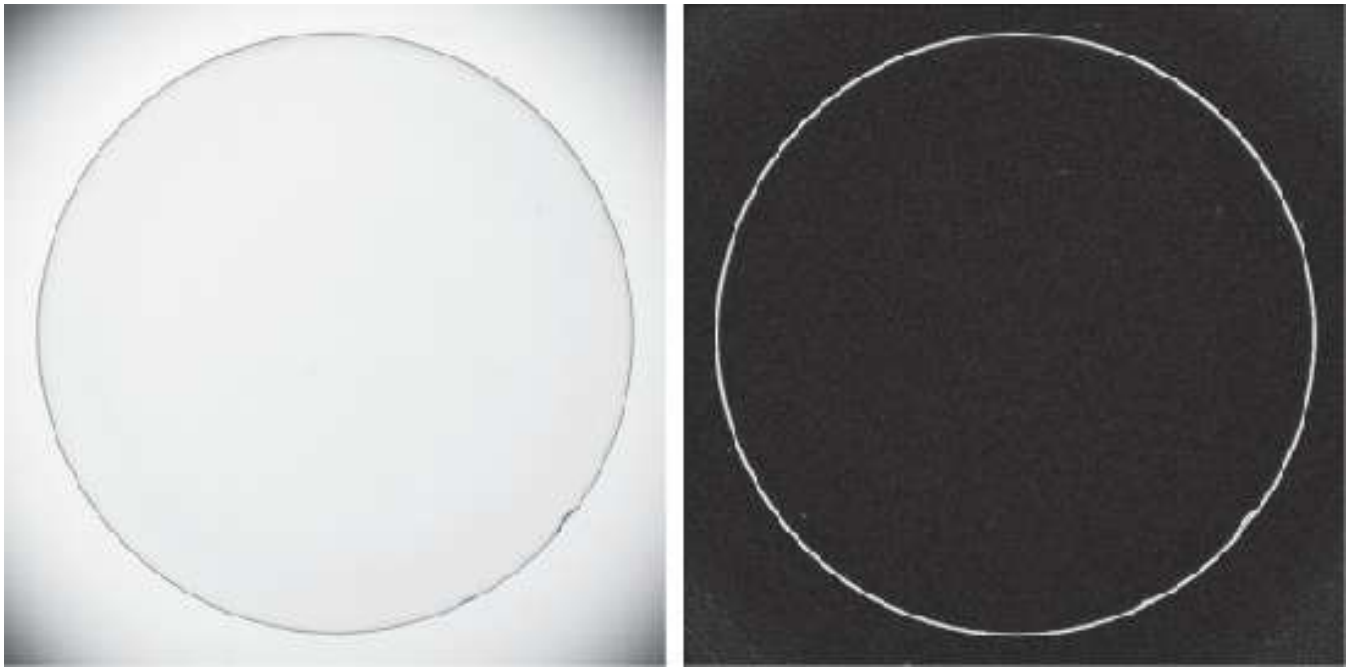
First Derivatives
The Gradient



a b

FIGURE 3.51 (a) Image of a contact lens (note defects on the boundary at 4 and 5 o'clock). (b) Sobel gradient. (Original image courtesy of Perceptics Corporation.)

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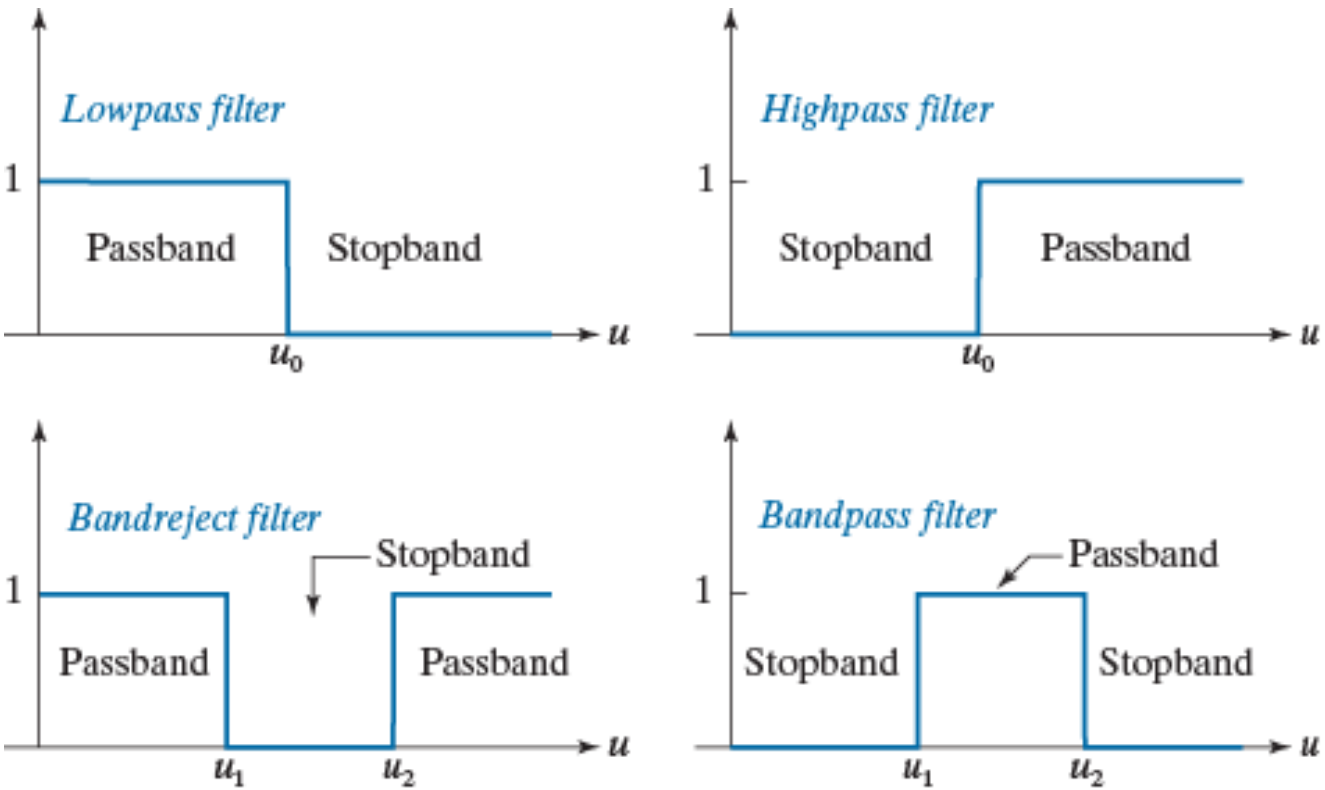
Filters from LPF
HP, BP, BR (more in Ch4)



a b
c d

FIGURE 3.52 Transfer functions of ideal 1-D filters in the frequency domain (u denotes frequency). (a) Lowpass filter. (b) Highpass filter. (c) Bandreject filter. (d) Bandpass filter. (As before, we show only positive frequencies for simplicity.)

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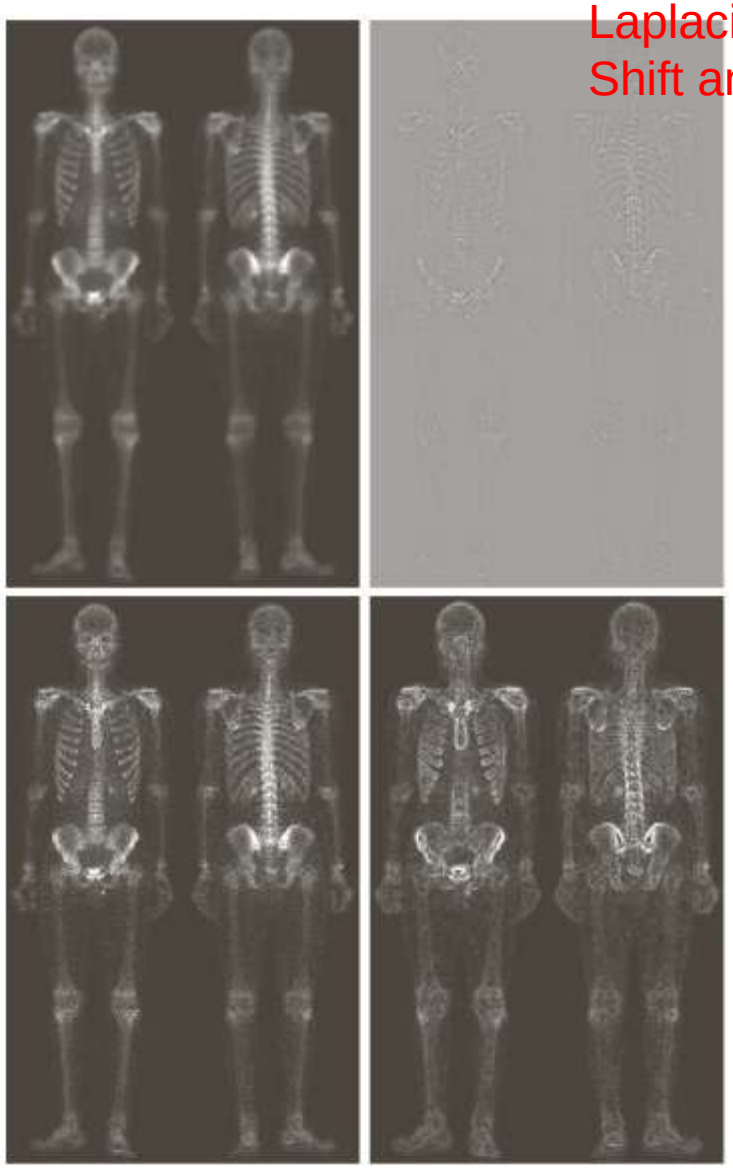
Combining Spatial Enhancement Methods

Laplacian

Shift and rescale to [0, L-1]



a b
c d

FIGURE 3.57 (a) Image of whole body bone scan. (b) Laplacian of (a). (c) Sharpened image obtained by adding (a) and (b). (d) Sobel gradient of image (a). (Original image courtesy of G.E. Medical Systems.)

A+B

Sobel

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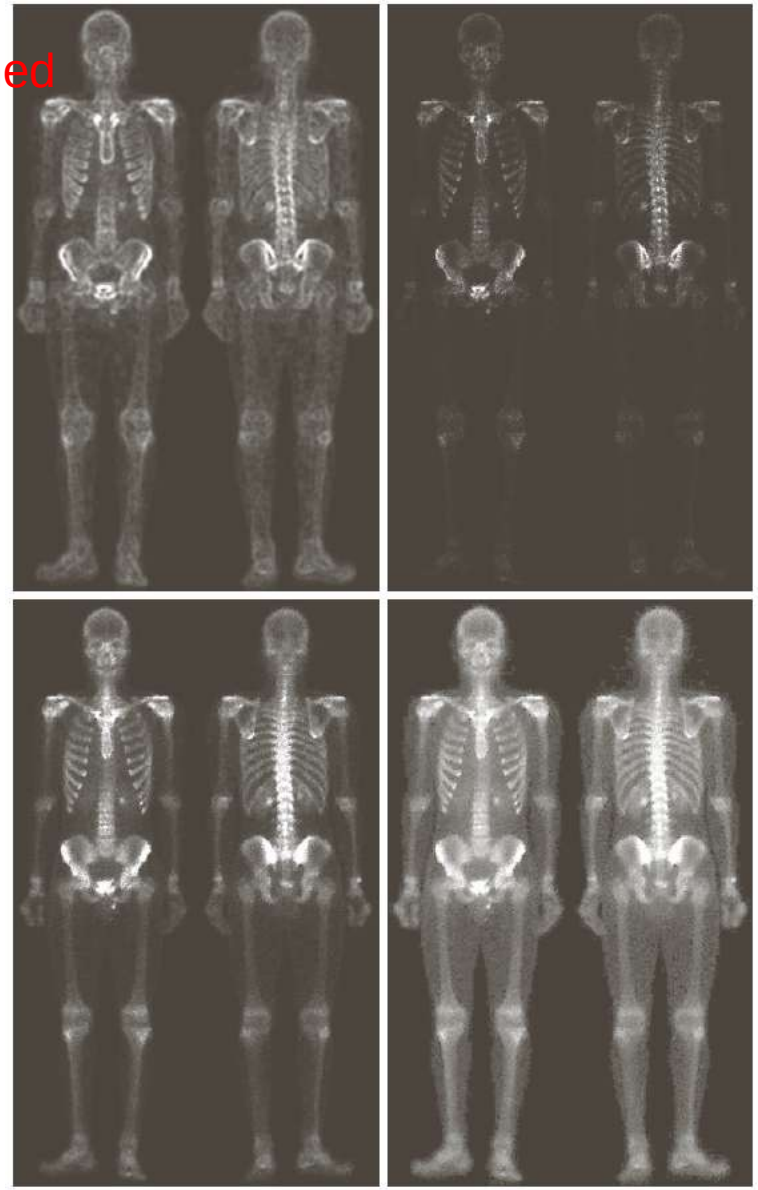
Combining Spatial Enhancement Methods

Smoothed sobel



e f
g h

FIGURE 3.57 *(Continued)* (e) Sobel image smoothed with a 5×5 box filter. (f) Mask image formed by the product of (b) and (e). (g) Sharpened image obtained by the adding images (a) and (f). (h) Final result obtained by applying a power-law transformation to (g). Compare images (g) and (h) with (a). (Original image courtesy of G.E. Medical Systems.)

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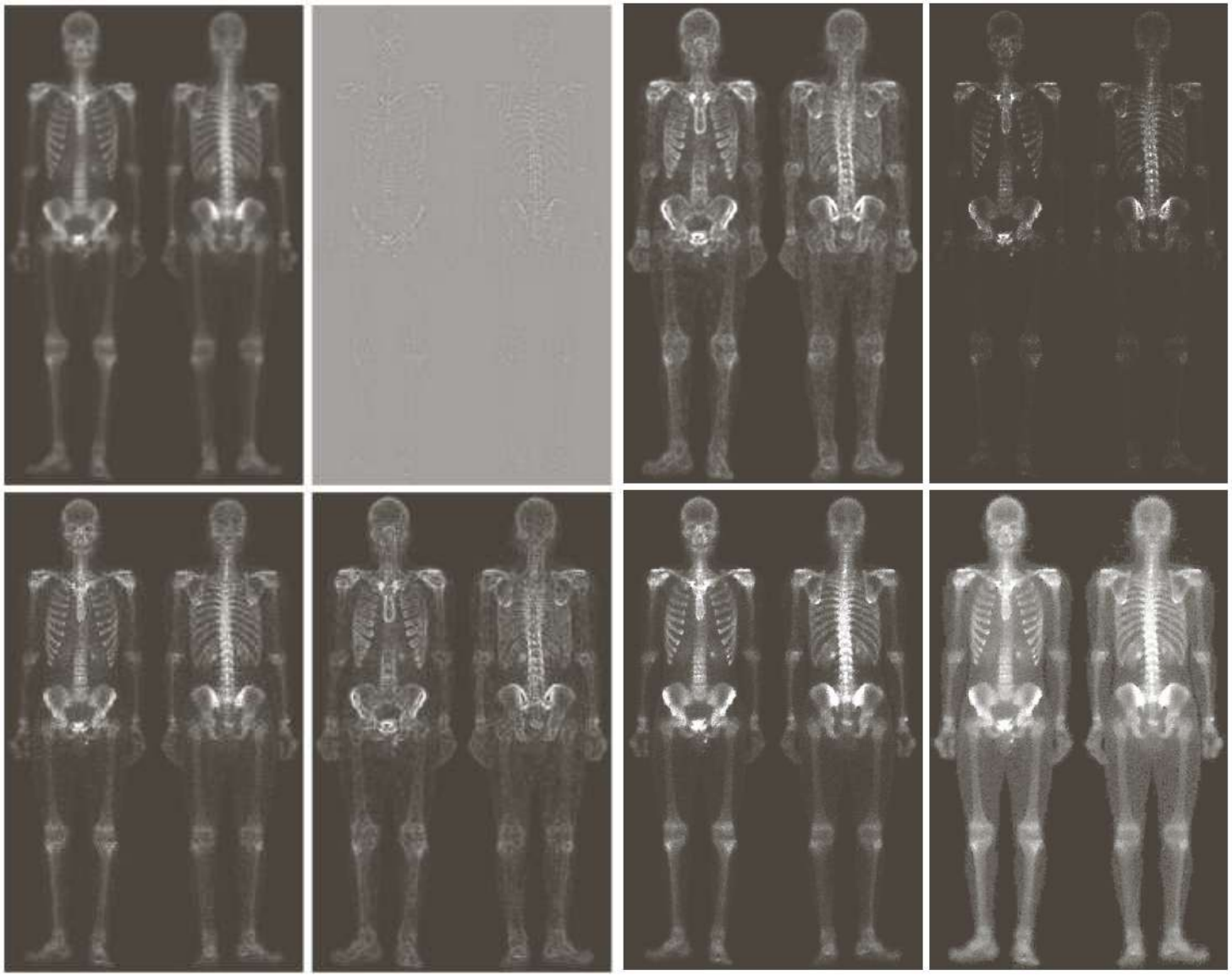
Combining Spatial Enhancement Methods



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